# Computer Networks

Autumn 2025

葛俊锋 13995529630

# 计算机网络

2025年秋季

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- Introduction: course description ( Why to learn? What to learn? How to learn?)
- Answers to frequently asked questions






- 简介：课程描述（为什么要 学习? 学习什么？如何学习？）
- 常见问题解答

# Textbook

1. Computer Networking: A Top Down Approach, 8th ed., Pearson/Addison Wesley
2. 计算机网络:自顶向下方法，原书第8版，机械工业出版社
   http://gaia.cs.umass.edu/kurose_ross/ppt.htm
3. 何顶新，郑定富，胡晓娅. *计算机网络及应用*，北京：化学工业出版社，2011.06

# Course Grading

- Assignments (experiment + homework ）+ attendance 30%
- Final 70%



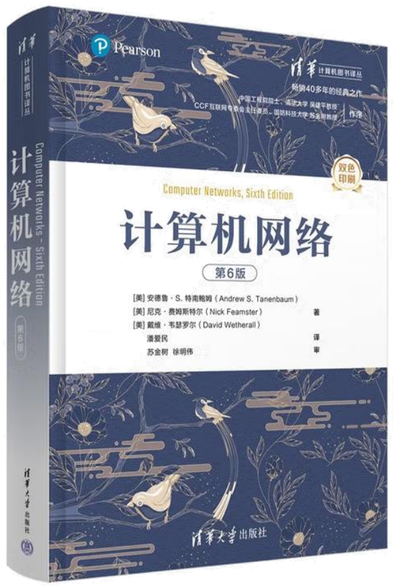
# 教材

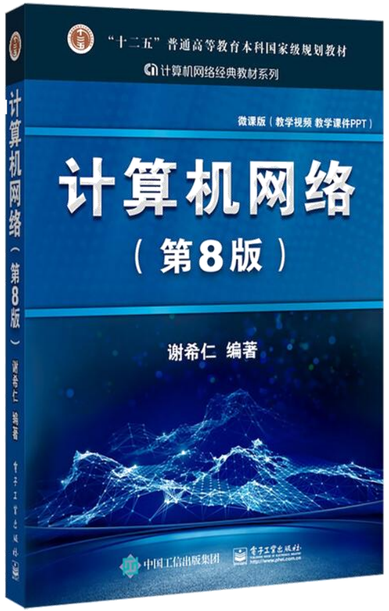
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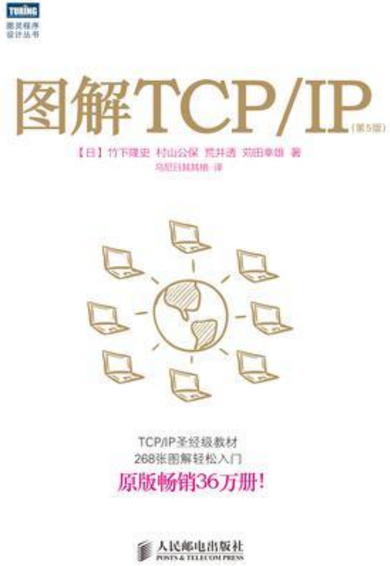
# 课程评分

- 作业 (实验 + 作业 )+ 出勤 30%
- 期末考试 70%

# Chapter 1: introduction



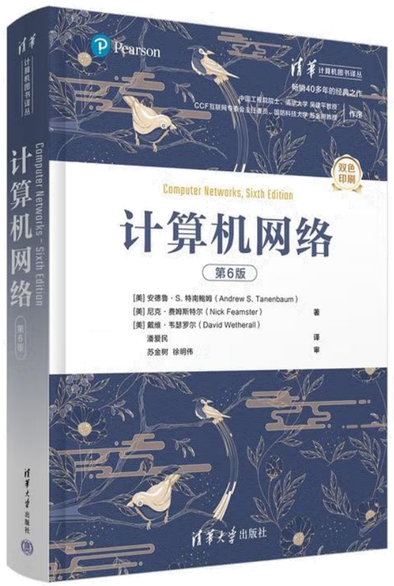



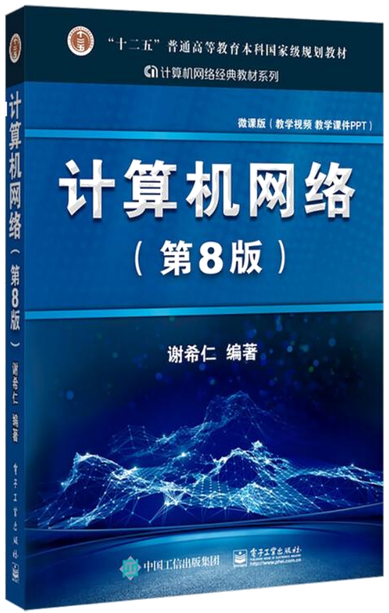

Andrew S.Tanenbaum, David J.Wet

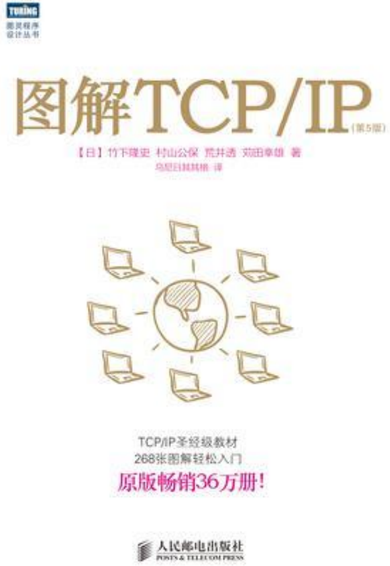
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# 第一章：引言




Andrew S.Tanenbaum, David J.Wet

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# Chapter 1: introduction

## Overview:

- Terminology and Concepts related to
  " Computer Network"
  - ✓ use Internet as example
- What's a protocol?
- Protocol layers and their service models
- Internet history

## Keypoints:

- Protocol and three essential factors
- Protocol layers and layered architecture

## Difficulties:

- Connection-oriented and connectionless
- Circuit switching and packet switching
- Delay and how to calculate the delay



# 第一章：引言

## 概述：

- 术语和
  与"网络"相关的概念
  "计算机网络"
  - ✓ 以互联网为例
- 什么是协议?
- 协议层及其服务模型
- 互联网历史

## 要点：

- 协议和三个基本因素
- 协议层和分层架构

## 困难：

- 面向连接和无连接
- 电路交换和分组交换
- 延迟和如何计算延迟

# Chapter 1: roadmap

1.1 What is the Internet?

1.2 Network edge

- hosts, access network, physical media

1.3 Network core

- packet switching, circuit switching, network structure

1.4 Delay & loss in packet-switched networks

1.5 Protocol layers, Service models

1.6* Internet history



# 第一章：路线图

1.1 什么是互联网?

1.2 网络边缘

- 主机、接入网、物理媒体

1.3 网络核心

- 分组交换、电路交换、网络结构

1.4 延迟与丢失 - 包交换网络

1.5 协议层与服务模型

1.6* 互联网历史

# What's the Internet: "nuts and bolts" view



Billions of connected computing *devices*:

- *hosts* = *end systems*
- running *network apps* at Internet's "edge"



*Packet switches*: forward packets (chunks of data)

- *routers, switches*

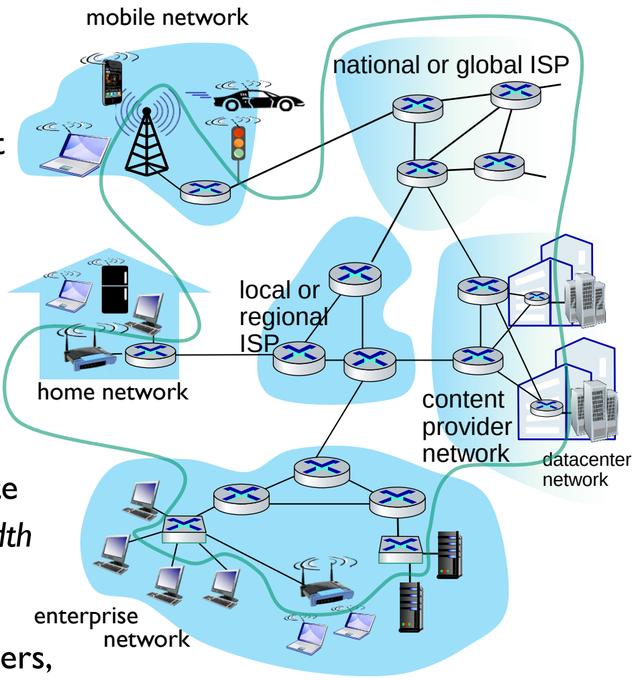


*Communication links*

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*

*Networks*

- collection of devices, routers, links: managed by an organization






# 什么是互联网：“螺丝钉和螺栓”视角



数十亿个互联的计算设备：

- 主机 = 终端系统
- 运行网络应用程序在互联网的“边缘”



分组交换机: 转发数据包（数据块）

- 路由器，交换机

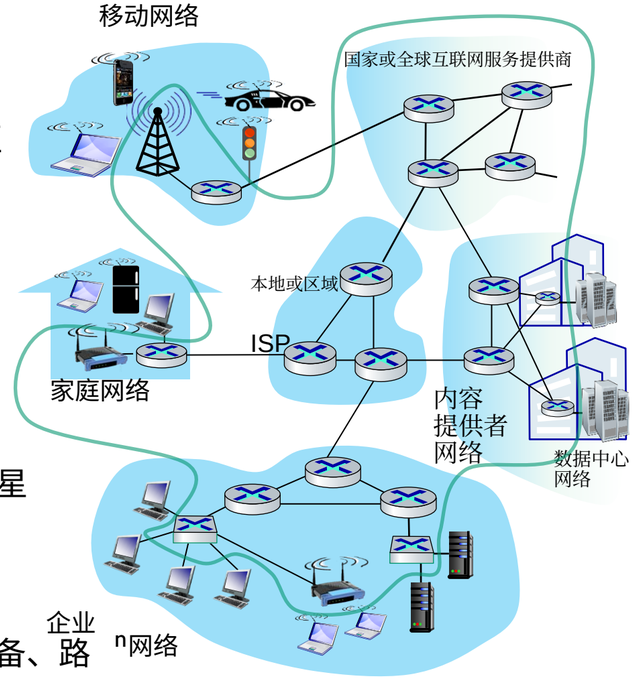


通信链路

- 光纤，铜线，无线电，卫星
- 传输速率：带宽

网络

- 由一个组织管理的一组设备、路由器和链路



# "Fun" Internet-connected devices


Amazon Echo


Internet refrigerator


IP picture frame


Pacemaker & Monitor


scooters


cars


bikes


Security Camera


Slingbox: remote control cable TV


Internet phones


Fitbit


AR devices


Tweet-a-watt: monitor energy use



# "有趣的"互联网连接设备


亚马逊回声


互联网冰箱


IP相框


起搏器与监控器


踏板车


cars


自行车


安全摄像头


Slingbox：远程控制有线电视


互联网电话

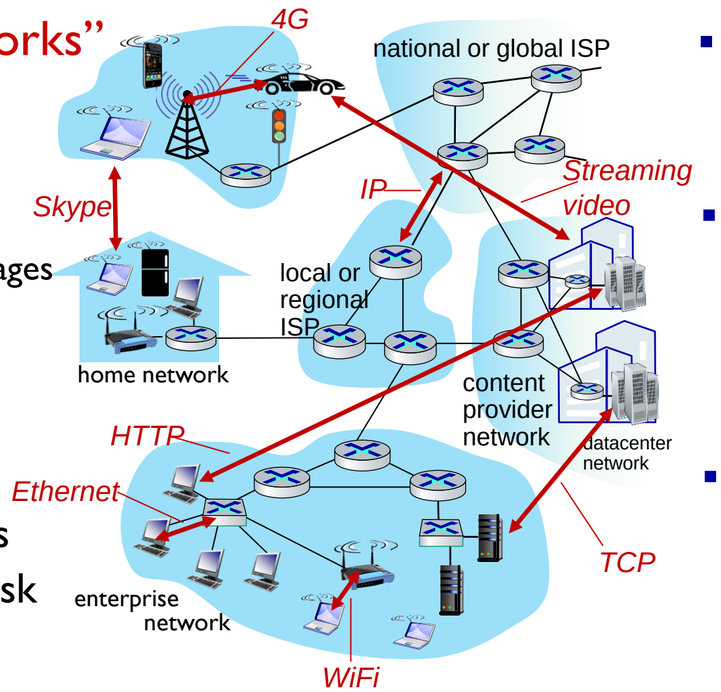

Fitbit


AR设备


推文-a-watt: 监控能源使用

# What's the Internet: "nuts and bolts" view

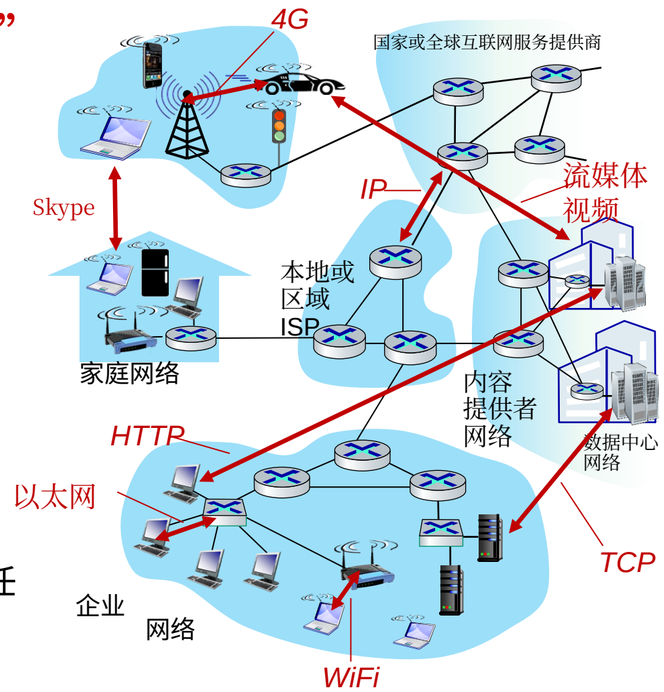
- *Internet:* "network of networks"
  - Interconnected ISPs
- *protocols* are *everywhere*
  - control sending, receiving of messages
  - e.g., HTTP (Web), TCP, IP, WiFi, streaming video, Skype, 4G, Ethernet
- *Internet standards*
  - RFC: Request for Comments
  - IETF: Internet Engineering Task Force





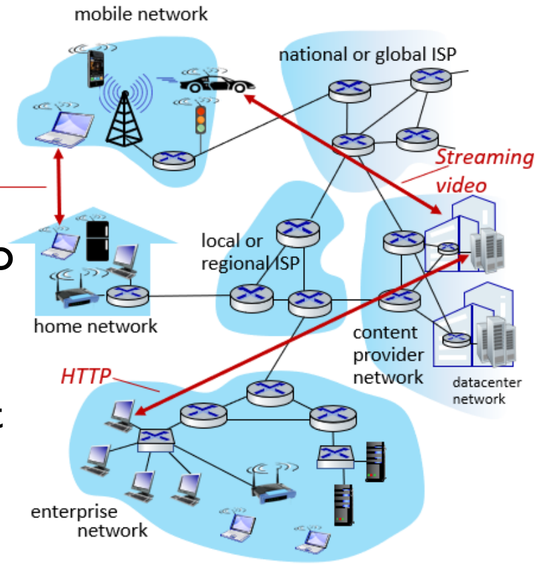

# 什么是互联网：“螺丝钉和螺栓”视角

- 互联网：“网络中的网络”
  - 互联的互联网服务提供商
- 协议无处不在
  - 控制消息的发送和接收
  - 例如，HTTP（网页），TCP，IP，WiFi，流媒体视频，Skype，4G，以太网
- 互联网标准
  - RFC：请求评论
  - 互联网工程任务组：互联网工程任务组



# What's the Internet: a service view

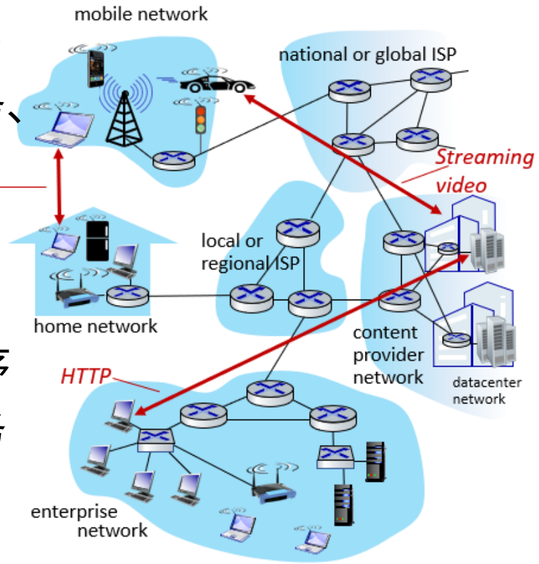
- *Infrastructure* that provides services to applications:
  - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, …
- provides *programming interface* to distributed applications:
  - "hooks" allowing sending/receiving apps to "connect" to, use Internet transport service
  - provides service options, **analogous to postal service**






# 什么'是互联网：一种服务视图

- 基础设施为应用提供服务的：
  - 网页、流媒体视频、多媒体远程会议、电子邮件、游戏、电子商务、社交媒体、互联家电，...
- 为分布式应用提供编程接口：
  - "钩子"允许发送/接收应用程序"连接"到并使用互联网传输服务
  - 提供服务选项，**类似于邮政服务**



# Chapter 1: roadmap

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1.2 Network edge

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1.3 Network core

- packet switching, circuit switching, network structure

1.4 Delay & loss in packet-switched networks

1.5 Protocol layers, Service models

1.6* Internet history



# 第一章：路线图

1.1 什么是互联网?

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- 主机、接入网、物理介质

1.3 网络核心

- 分组交换、电路交换、网络结构

1.4 延迟与丢失 - 包交换网络

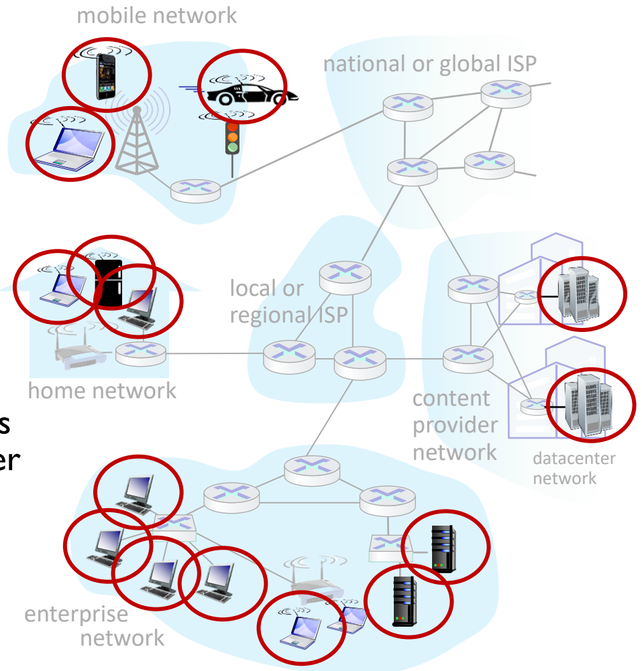
1.5 协议层与服务模型

1.6* 互联网历史

# A closer look at network structure:

## Network edge:

- hosts (End systems):
  - Run application programs
  - e.g. Web, email
  - At "edge of network"
  - clients and servers
- Client/server model
  - Client host requests, receives service from always-on server
  - E.g. Web browser/server; email client/server
- Peer-to-peer model
  - Minimal (or no) use of dedicated servers
  - E.g. Skype, BitTorrent, KaZaA

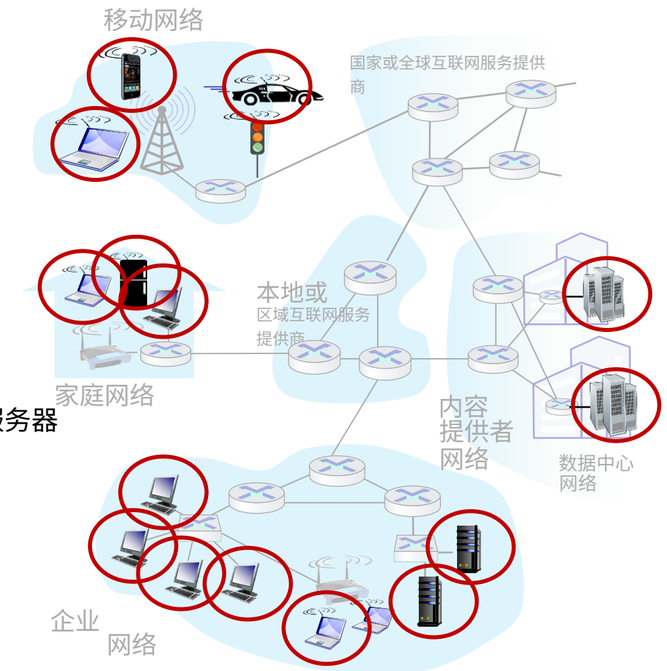





# 深入网络结构：

## 网络边缘：

- 主机 (终端系统):
  - 运行应用程序
  - 例如 Web、电子邮件
  - 在“网络边缘”
  - 客户端和服务器
- 客户端/服务器模型
  - 客户端主机请求，从始终在线的服务器获取服务
  - 例如：网页浏览器/服务器；电子邮件客户端/服务器
- 对等模型
  - 极少（或不）使用专用服务器
  - 例如 Skype、BitTorrent、KaZaA



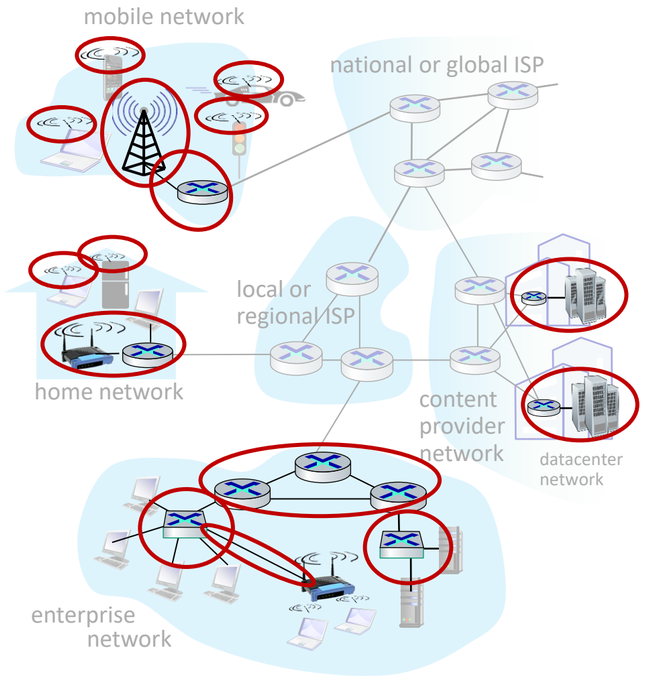
# A closer look at network structure:

## Network edge:

- hosts: clients and servers
- servers often in data centers

## Access networks, physical media:

- wired, wireless communication links





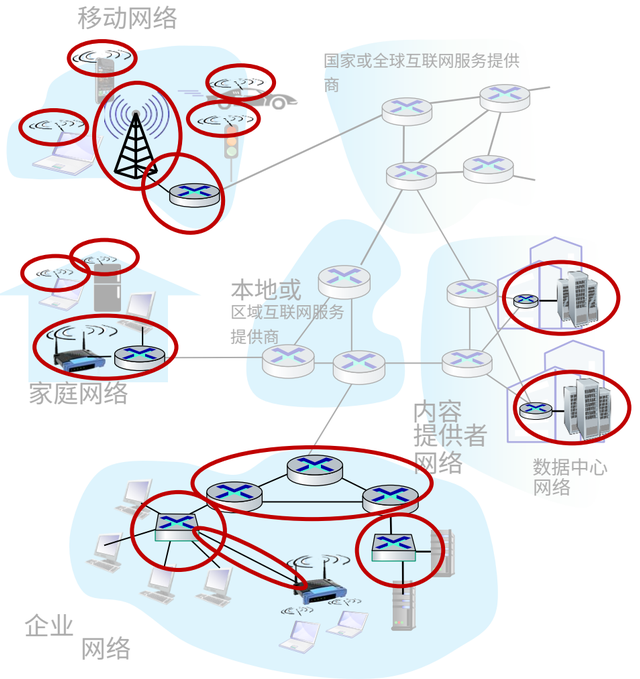

# 深入网络结构：

## 网络边缘：

- 主机：客户端和服务器
- 服务器通常位于数据中心

## 接入网络，物理介质：

- 有线、无线通信链路



# A closer look at network structure:

## Network edge:

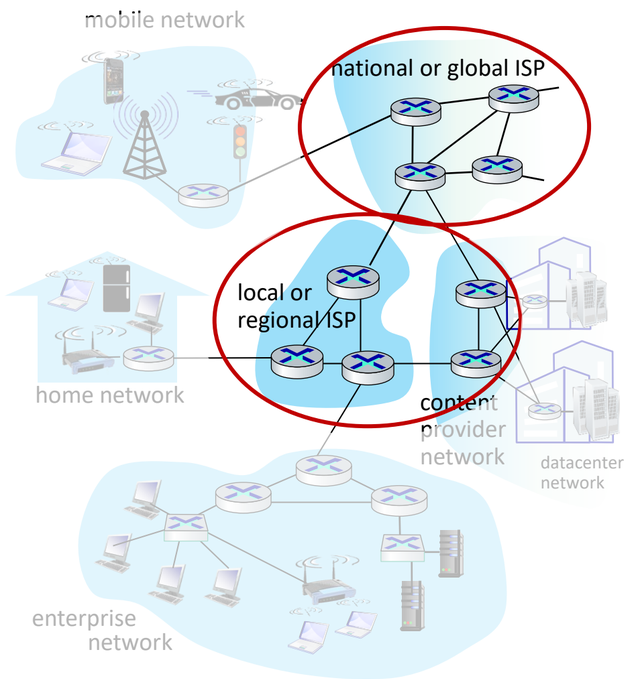
- hosts: clients and servers
- servers often in data centers

## Access networks, physical media:

- wired, wireless communication links

## Network core:

- interconnected routers
- network of networks






# 深入网络结构：

## 网络边缘：

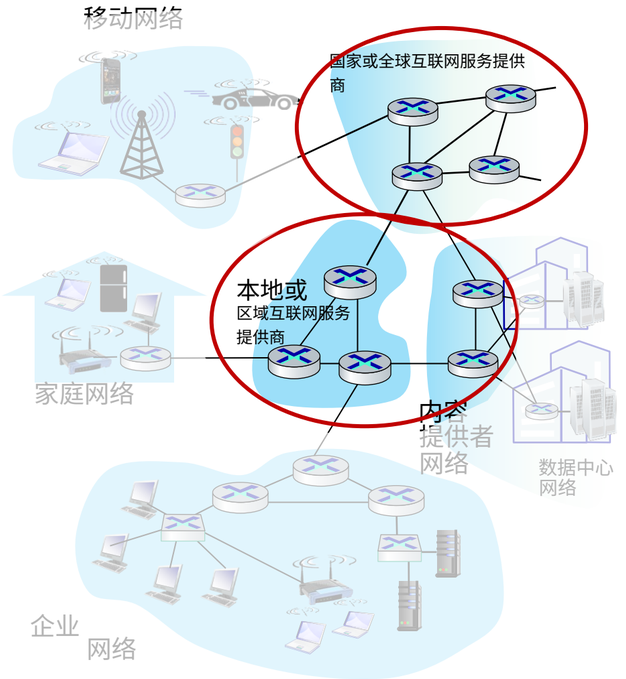
- 主机：客户端和服务器
- 服务器通常位于数据中心

## 接入网络，物理介质：

- 有线、无线通信链路

## 网络核心：

- 相互连接的路由器
- 网络中的网络



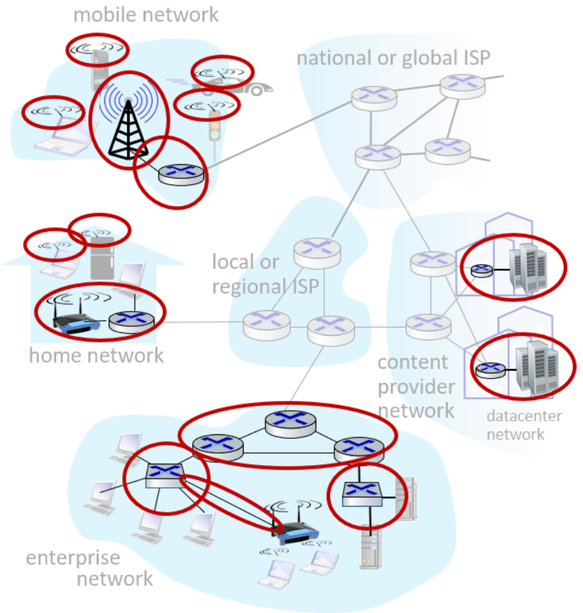
# Access networks and physical media

*Q: How to connect end systems to edge router?*

- residential access nets
- institutional access networks (school, company)
- mobile access networks (WiFi, 4G/5G)

*What to look for:*

- bandwidth (bits per second) of access network?
- shared or dedicated?





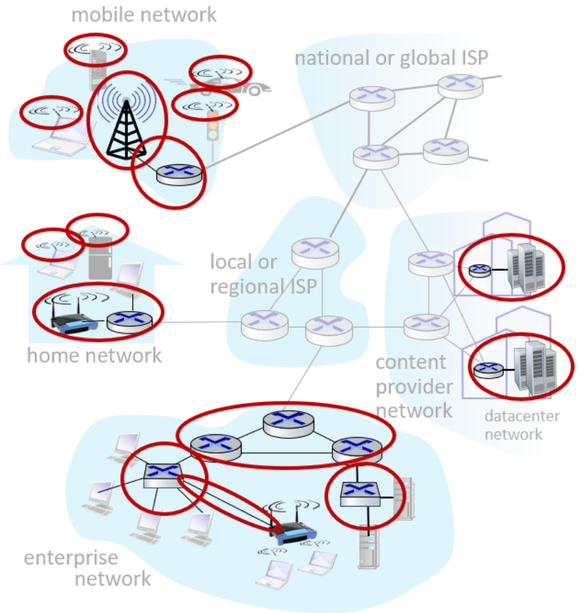

# 接入网络和物理媒体

Q：如何将终端系统连接到边缘路由器？

- 住宅接入网
- 机构接入网络（学校、公司）
- 移动接入网络（WiFi、4G/5G）

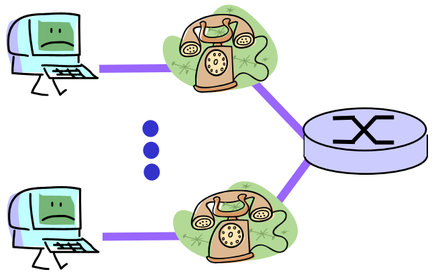
需要关注的事项：

- 接入网的带宽（比特每秒）？
- 共享还是专用？



# Residential Access: Point to Point Access

- Dialup via modem
  - Up to 56Kbps direct access to router (often less)
  - Can't surf and phone at same time: can't be "always on"

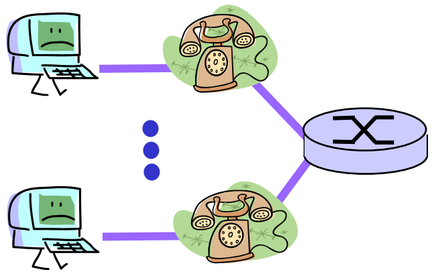


- ADSL: asymmetric digital subscriber line
  - Up to 1 Mbps upstream (today typically < 256 kbps)
  - Up to 8 Mbps downstream (today typically < 1 Mbps)
- HFC: hybrid fiber coax
  - Asymmetric: up to 30Mbps downstream, 2 Mbps upstream
  - Is shared broadcast medium



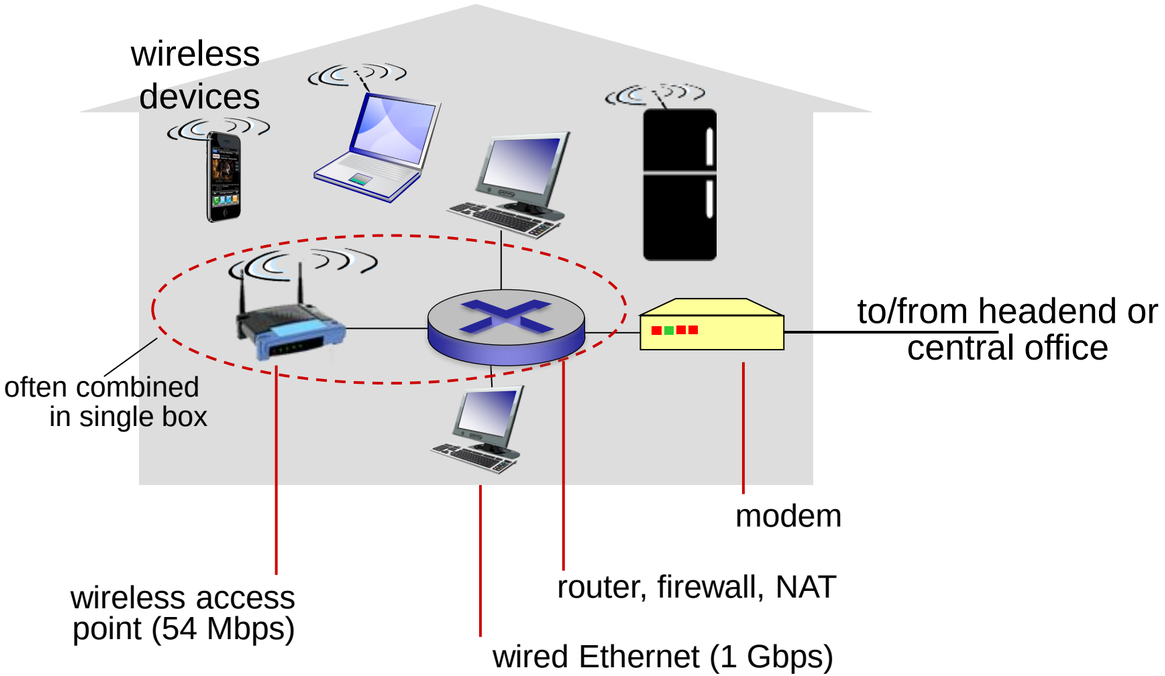
# 住宅接入：点对点接入

- 通过调制解调器拨号
  - 最高56Kbps直接访问路由器（通常更低）
  - 不能同时上网和打电话：不能实现"永远在线"



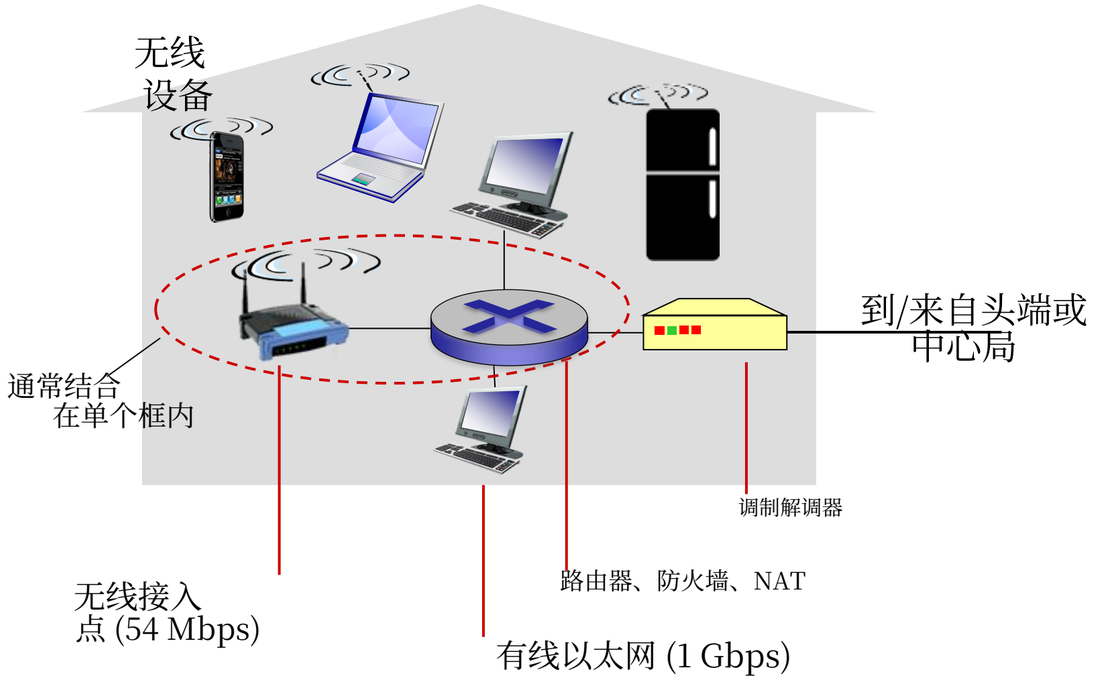
- ADSL：非对称数字用户线路
  - 上行速率最高1 Mbps（目前通常为 < 256 kbps）
  - 下行速率最高8 Mbps（目前通常为 < 1 Mbps）
- HFC：混合光纤同轴电缆
  - 非对称：下行最高 30Mbps，上行 2Mbps
  - 共享广播介质

# Access network: home network

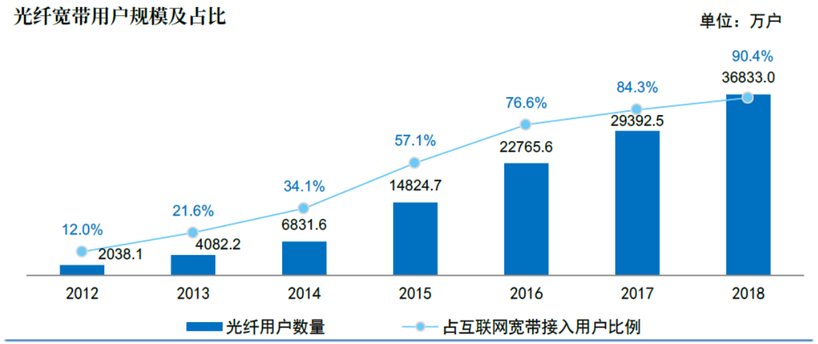




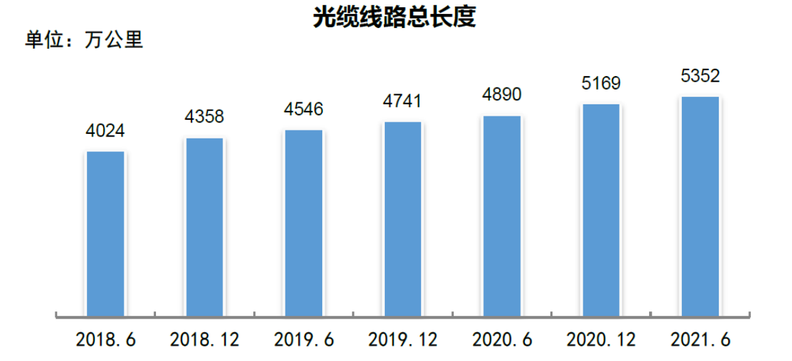

# 接入网络：家庭网络



# Access network: home network

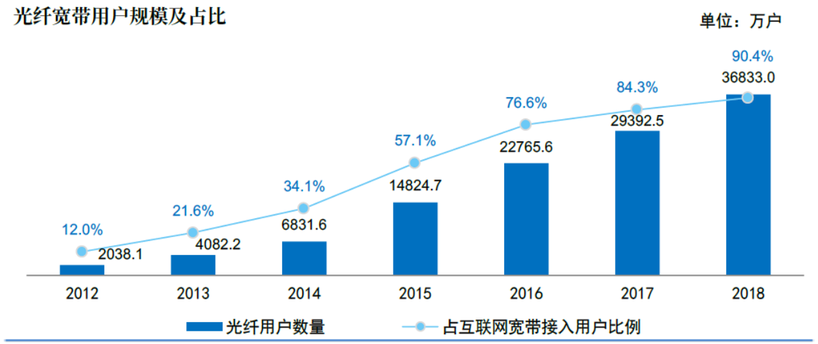



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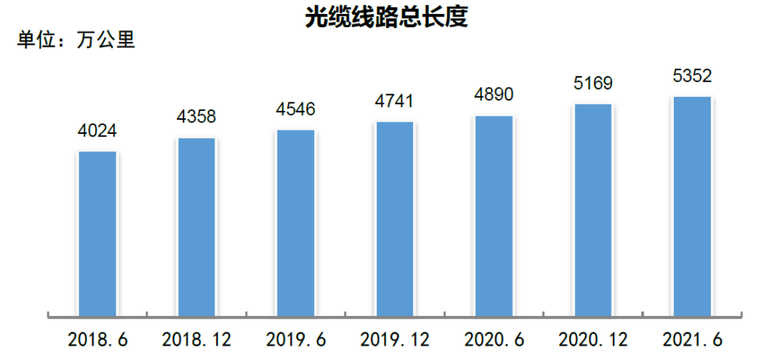



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# 接入网络：家庭网络

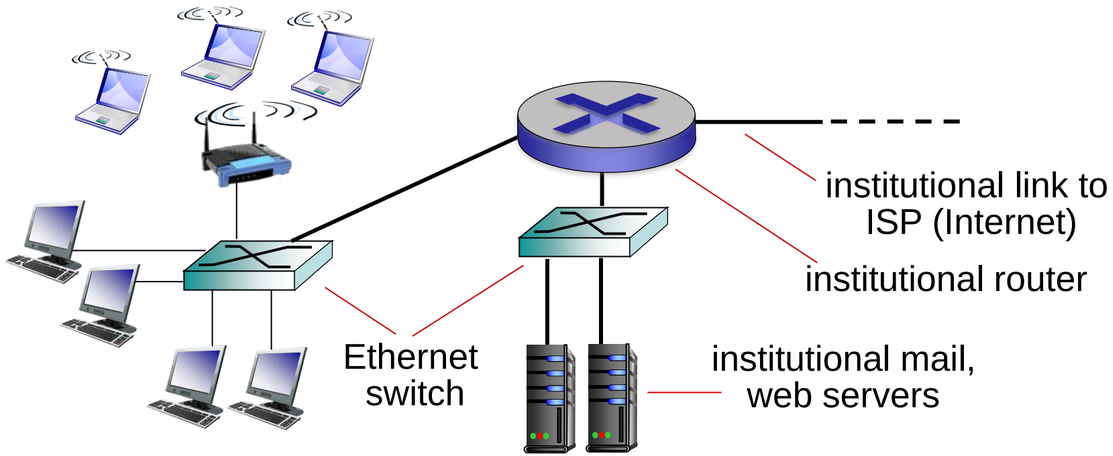



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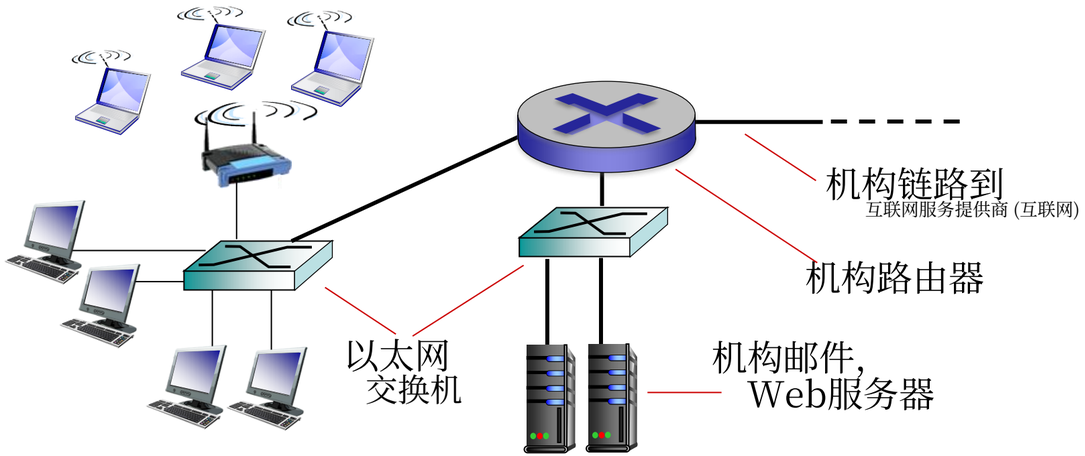
# Enterprise access networks (Ethernet)




- typically used in companies, universities, etc.
- 10 Mbps, 100Mbps, 1Gbps, 10Gbps transmission rates
- today, end systems typically connect into Ethernet switch



# 企业接入网络（以太网）



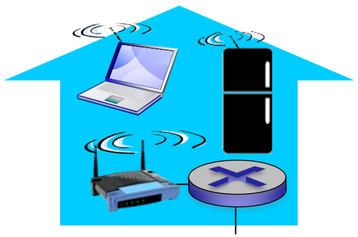

▪通常用于公司、大学等场所。▪ 10 Mbps、100 Mbps、1 Gbps、10 Gbps 的传输速率▪ 如今，终端系统通常连接到以太网交换机

# Wireless access networks

- shared *wireless* access network connects end system to router
  - via base station aka "access point"

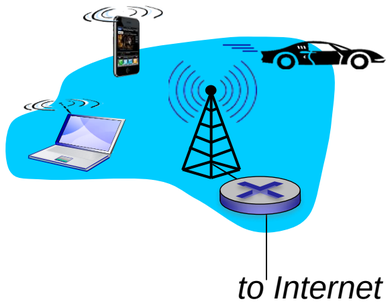
*wireless LANs:*

- within building (100 ft.)
- 802.11b/g/n/ac/ax (WiFi): 11, 54, 450 Mbps transmission rate




wide-area wireless access

- provided by telco (cellular) operator, 10's km
- between 1 and 10 Mbps
- 3G, 4G: LTE, 5G



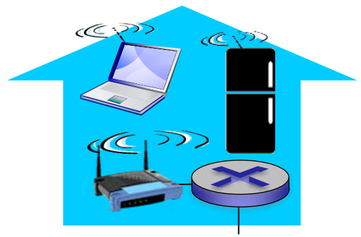



# 无线接入网络

- 共享无线接入网络将终端系统连接到路由器
  - 通过 基站 又名 "接入点"

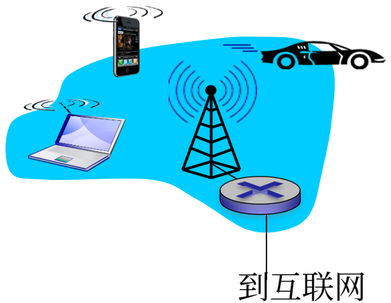
无线局域网：

- 建筑物内部（约30米）
- 802.11b/g/n/ac/ax (WiFi): 11, 54, 450 Mbps 传输速率




广域无线接入

- 由电信运营商提供，10's km
- 1到10 Mbps之间
- 3G, 4G: LTE,5G



# Physical media

- bit: propagates between transmitter/receiver pairs
- physical link: what lies between transmitter & receiver
- guided media:
  - signals propagate in solid media: copper, fiber, coax
- unguided media:
  - signals propagate freely, e.g., radio

*twisted pair (TP)*

- two insulated copper wires
  - Category 5: 100 Mbps, 1 Gbps Ethernet
  - Category 6: 10Gbps



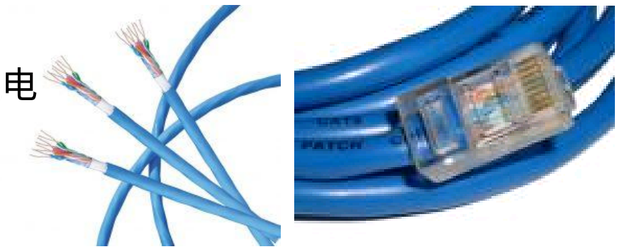


# 物理媒体

- bit: 在发射器/接收器对之间传播
- 物理链路：发射器与接收器之间是什么
- 导向媒体：
  - 信号在固体媒体中传播：铜线、光纤、同轴电缆
- 无导向媒体：
  - 信号可以自由传播，例如无线电

双绞线 (TP)

- 两根绝缘铜线
  - 五类线缆：100 Mbps，1 Gbps 以太网
  - 六类线缆：10Gbps



# Physical media: coax, fiber



*coaxial cable:*

- two concentric copper conductors
- bidirectional
- broadband:
  - multiple channels on cable
  - HFC





*fiber optic cable:*

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
  - high-speed point-to-point transmission (e.g., 10' s-100' s Gbps transmission rate)
- low error rate:
  - repeaters spaced far apart
  - immune to electromagnetic noise

长飞、亨通光电、烽火



中国光谷已成长为中国最大的光纤光缆、光电器件生产基地，最大的光通信技术研发基地。



# 物理介质：同轴电缆、光纤



同轴电缆：

- 两根同心铜导体
- 双向
- 宽带：
  - 在同轴电缆上支持多个信道
  - HFC





光纤：

- 玻璃纤维传输光脉冲，每个脉冲代表一个比特
- 高速运行：
  - 高速点对点传输（例如，10' 秒-100' 秒千兆比特每秒传输速率）
- 低误码率：
  - 远距离间隔的中继器
  - 不受电磁干扰

长飞、亨通光电、烽火



中国光谷已成数据包长度中国最大的光 纤光缆、光电器件生产基地,最大 的光通信技术研发基地。

# Physical media: radio

- signal carried in electromagnetic spectrum
- no physical "wire"
- bidirectional
- propagation environment effects:
  - reflection
  - obstruction by objects
  - interference

*radio link types:*

- terrestrial microwave
  - e.g. up to 45 Mbps channels
- LAN (e.g., WiFi)
  - 54 Mbps
- wide-area (e.g., cellular)
  - 4G cellular: ~ 10 Mbps
- satellite
  - Kbps to 45Mbps channel (or multiple smaller channels)
  - 270 msec end-end delay
  - geosynchronous versus low altitude



# Phy无线媒体：无线电

- 信号在电磁频谱中传输
- 没有物理"线缆"
- 双向
- 传播环境影响因素：
  - 反射
  - 物体阻挡
  - 干扰

无线电链路类型：

- 地面微波
  - 例如，高达45 Mbps信道
- 局域网 (例如，WiFi)
  - 54 Mbps
- 广域(例如，蜂窝网络)
  - 4G蜂窝：~ 10 Mbps
- 卫星
  - Kbps至45Mbps信道（或多个较小信道）
  - 270毫秒 端到端延迟
  - 地球同步与低空

# Chapter 1: roadmap

1.1 What is the Internet?
1.2 Network edge
- hosts, access network, physical media

1.3 Network core
- packet switching, circuit switching, network structure

1.4 Delay & loss in packet-switched networks
1.5 Protocol layers, Service models
1.6* Internet history



# 第一章：路线图

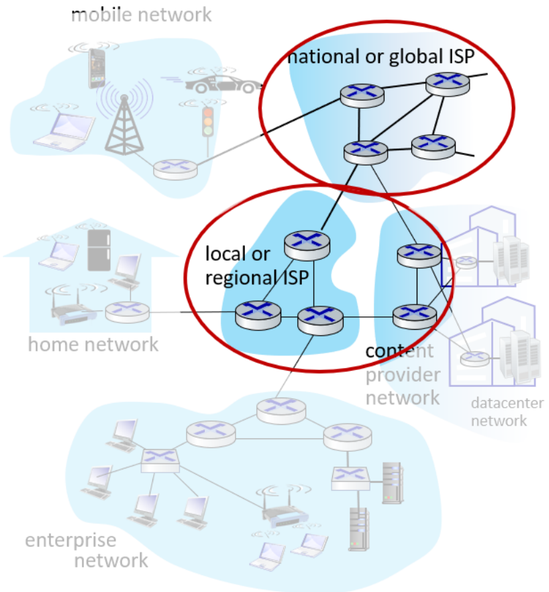
1.1 什么是互联网?
1.2 网络边缘
- 主机、接入网、物理媒体

1.3 网络核心
- 分组交换、电路交换、网络结构

1.4 延迟与丢失 - 包交换网络
1.5 协议层与服务模型
1.6* 互联网历史

# The network core

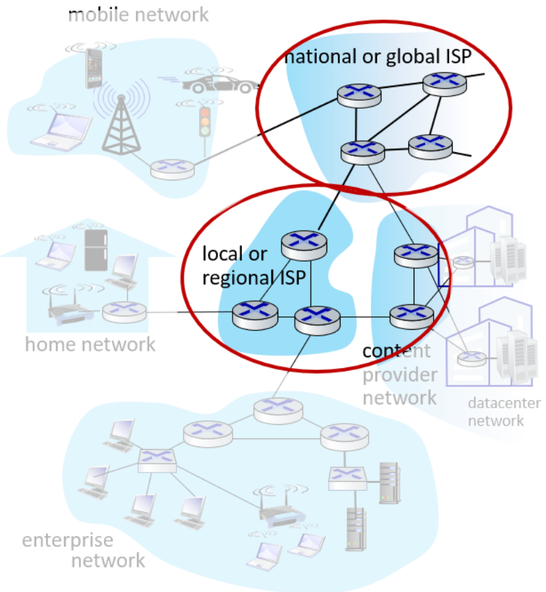
- mesh of interconnected routers
- *The fundamental question:* how is data transferred through net?
  - Circuit-switching: dedicated circuit per call: telephone net
  - Packet-switching: data sent through net in discrete "chunks"



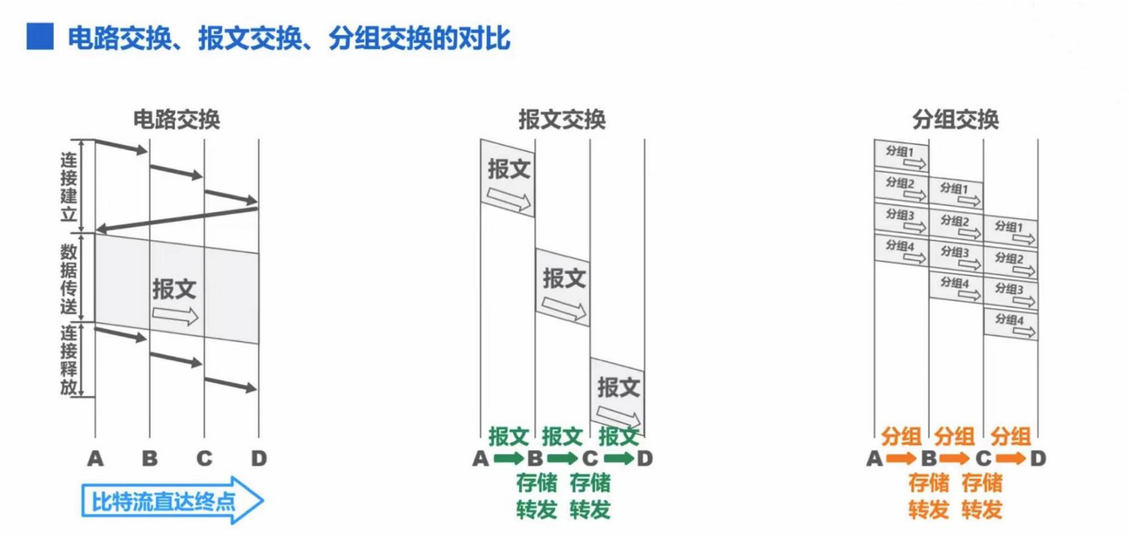



# 网络核心

- 互联路由器的网状结构
- 基本问题：： 数据如何通过网络传输?
  - 电路交换： 每个呼叫专用电路：电话网
  - 分组交换：数据通过网络以离散的“数据块”发送



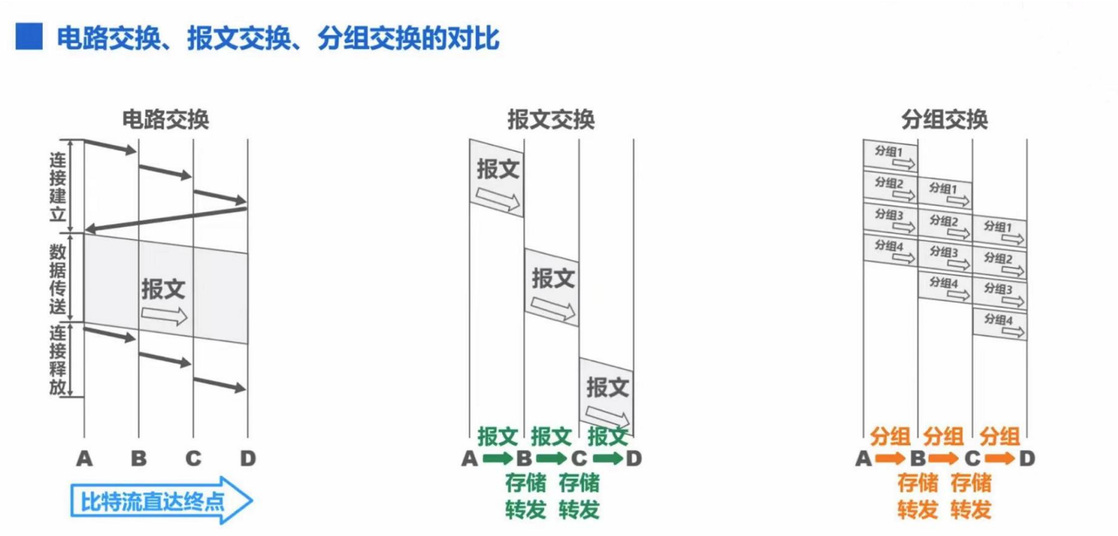
# The network core




Message switching and Packet-Switching: Store-and-Forward



# 网络核心

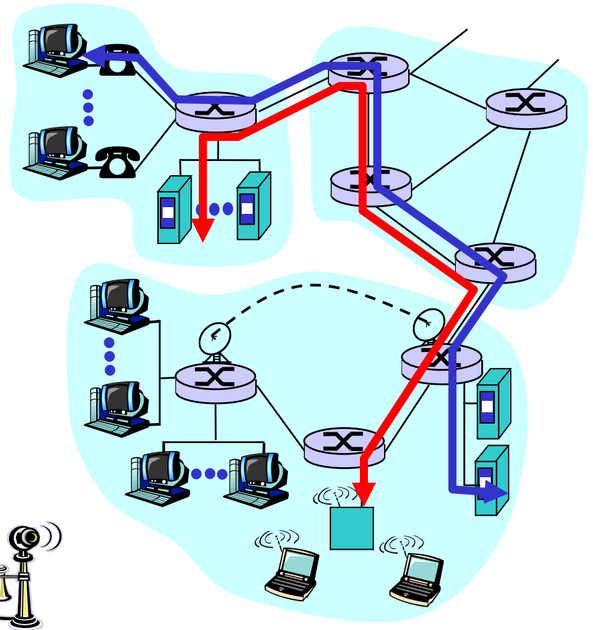


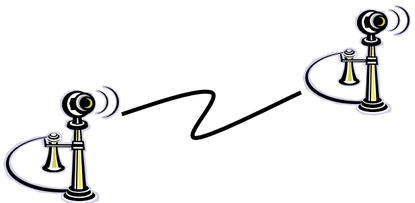

消息交换和分组交换：存储转发

# Circuit Switching

End-end resources reserved for "call"

- Link bandwidth, switch capacity
- Dedicated resources: no sharing
- Circuit-like (guaranteed) performance
- Call setup required



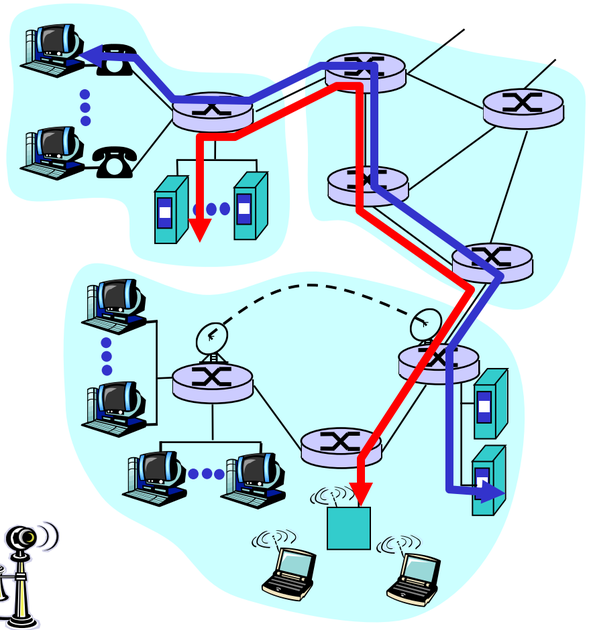


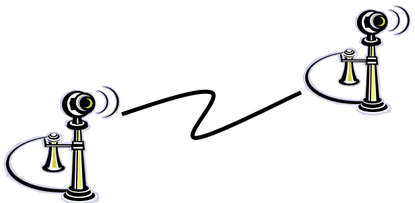


# 电路交换

端到端资源保留用于“呼叫”

- 链路带宽，交换机容量
- 专用资源：不共享
- 电路式（保证）性能
- 呼叫建立需要





# Circuit Switching

Network resources (e.g., bandwidth) divided into "pieces"

- Pieces allocated to calls
- Resource piece *idle* if not used by owning call *(no sharing)*

- Dividing link bandwidth into "pieces"
  - Frequency division
  - Time division







# 电路交换

网络资源(例如，带宽)被划分为"块"

- 分配给呼叫的片段
- 资源片若未被所属呼叫使用则空闲(不共享)

- 将链路带宽 分成"块"
  - 频分复用 时分复用

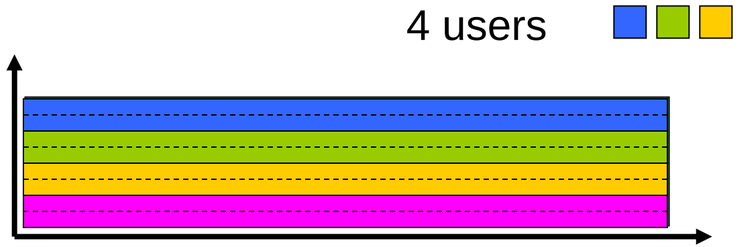


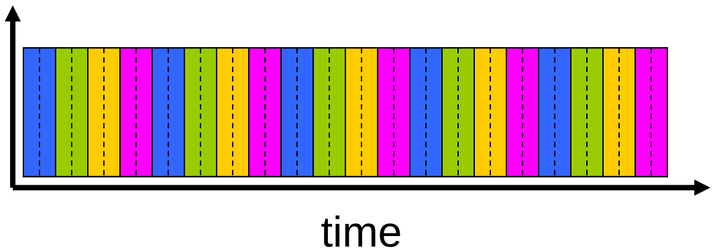


# Circuit switching: FDM versus TDM

Example:

4 users



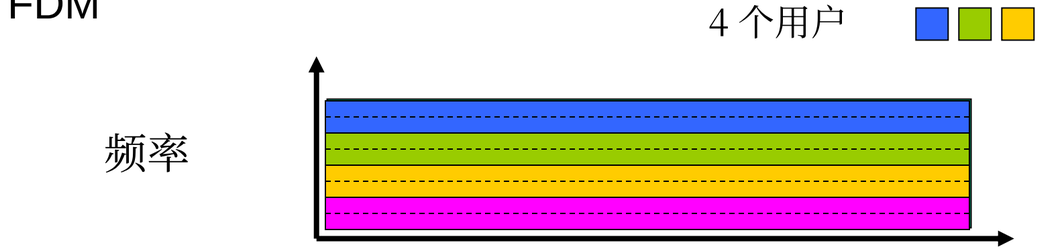



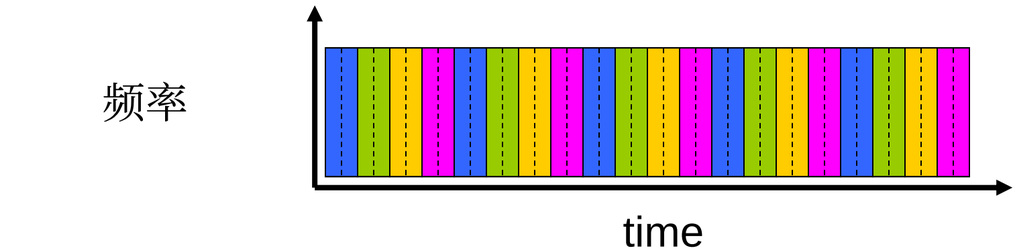



# 电路交换：FDM 与 TDM

示例：

4 个用户



# Numerical Example

- How long does it take to send a file of 640,000 bits from host A to host B over a circuit-switched network?
  - All links are 1.536 Mbps
  - Each link uses TDM with 24 slots
  - 500 msec to establish end-to-end circuit

Work it out!





# 数值示例

- 在电路交换网络中，从主机 A 发送 640,000 比特大小的文件到主机 B 需要多长时间?
  - 所有链路均为 1.536 Mbps
  - 每条链路使用 TDM 并有 24 个时隙
  - 500 建立端到端电路需要 毫秒

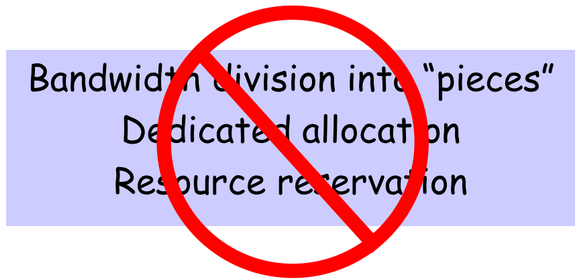
动起来！



# Network Core: Packet Switching

Each end-end data stream divided into *packets*

- User A, B packets *share* network resources
- Each packet uses full link bandwidth
- Resources used *as needed*




Resource contention:

- Aggregate resource demand can exceed amount available
- Congestion: packets queue, wait for link use
- Store and forward: packets move one hop at a time
  - Node receives complete packet before forwarding



# 网络核心：分组交换

每个端到端数据流被划分为<样式 id='1'> 数据包 </ 样式 >

- 用户 A、B 数据包共享网络资源

<样式 id='1'> ▪ </样式><样式 id='3'> 每个数据包使用完整链路带宽</样式>

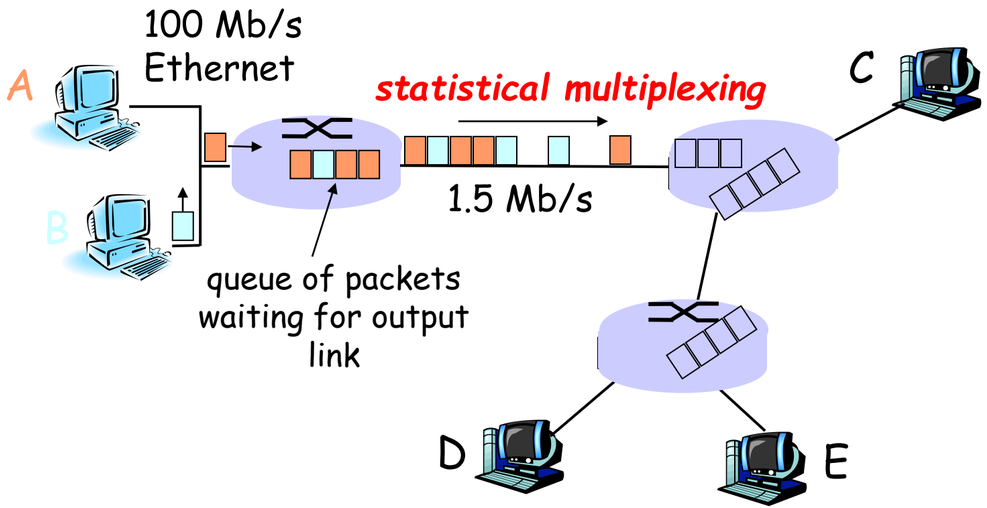
- 资源使用按需




资源争用：

- 聚合资源需求可能超过可用量
- 拥塞：数据包排队，等待链路使用
- <样式 id='1'>存储转发：</样式>数据包逐跳移动
  - 节点在转发前接收完整数据包

# Packet Switching: Statistical Multiplexing

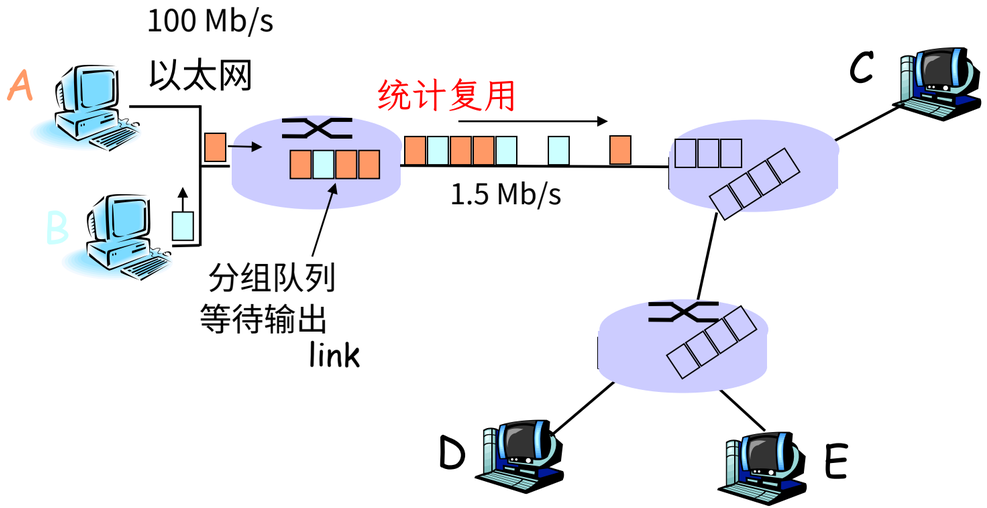



Sequence of A & B packets does not have fixed pattern, shared on demand ***statistical multiplexing***

TDM: each host gets same slot in revolving TDM frame



# 分组交换：统计复用



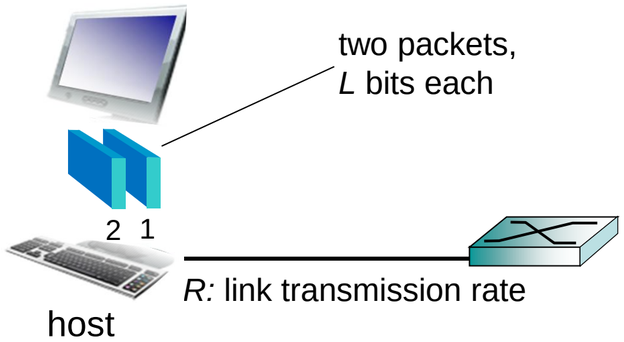

A与B数据包的序列没有固定模式，按需共享 统计复用 TDM：

每个主机在旋转TDM帧中获得相同时隙

# Host: sends packets of data

host sending function:

- takes application message
- breaks into smaller chunks, known as *packets*, of length *L* bits
- transmits packet into access network at *transmission rate R*
  - link transmission rate, aka link *capacity, aka link bandwidth*



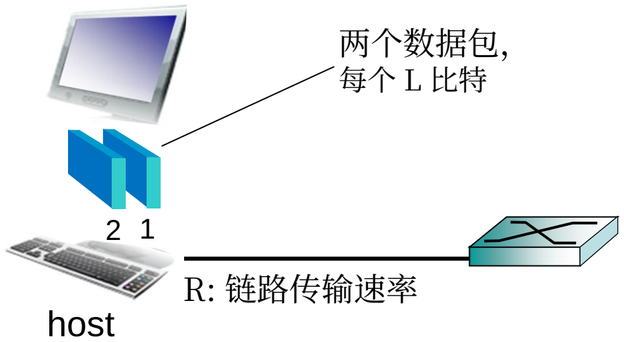

$$\text{packet transmission delay} = \text{time needed to transmit } L\text{-bit packet into link} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$



# 主机：发送数据包的数据

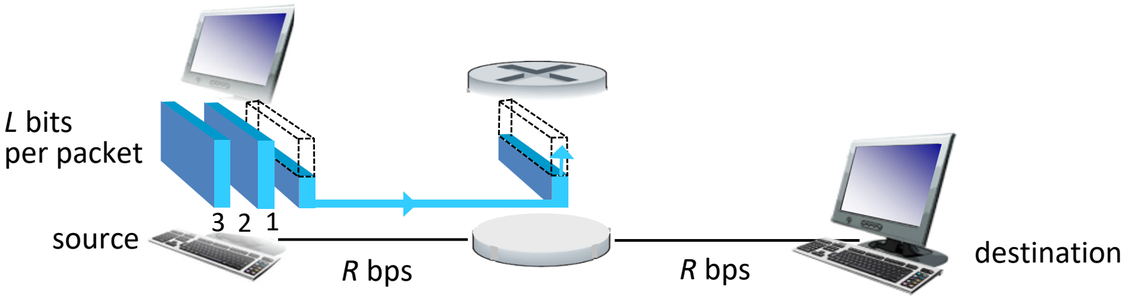
主机发送功能：

- 接收应用消息
- 将其拆分为更小的块，称为数据包，长度为*L*比特
- 以传输速率 R将数据包传输到接入网络
  - 链路传输速率，也称为链路容量，也称为链路带宽




数据包传输延迟时间，将 *L*-比特数据包传输到链路 *L* (比特) R(比特/秒) = =

# Packet-switching: store-and-forward




- Transmission delay: takes *L*/*R* seconds to transmit (push out) *L*-bit packet into link at *R* bps
- store and forward : entire packet must arrive at router before it can be transmitted on next link
- end-end delay = 2*L*/*R* (assuming zero propagation delay)

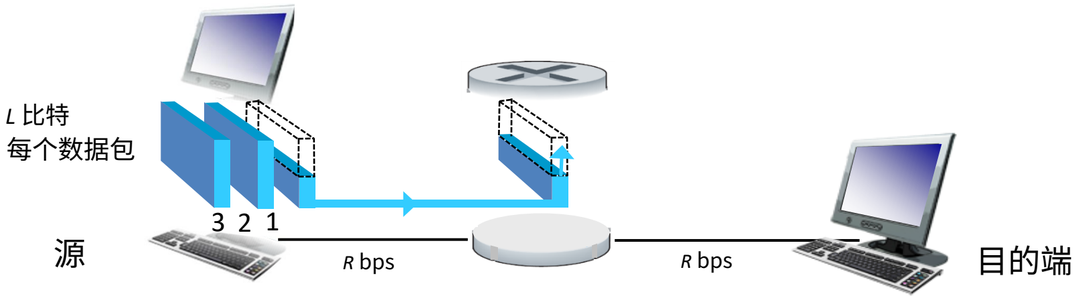
*one-hop numerical example:*

- *L* = 7.5 Mbits
- *R* = 1.5 Mbps
- one-hop transmission delay = 5 sec

more on delay shortly …



# 分组交换：存储转发




- 传输延迟：需要 *L*/*R* 秒将 *L*-比特数据包传输到链路，在 *R* bps
- 存储转发： 整个数据包必须先到达路由器，才能在下一个链路上传输
- 端到端延迟 = 2*L*/*R*(假设传播延迟为零)

单跳数值示例：

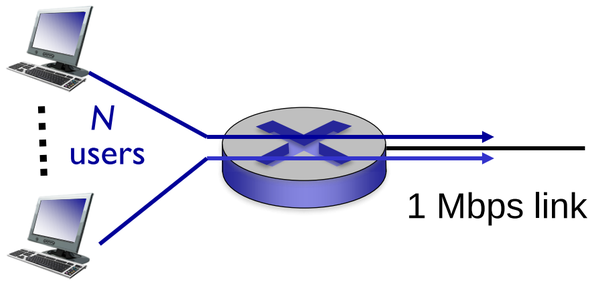
- 数据包长度 = 7.5 M比特
- 链路带宽 = 1.5 Mbps
- 单跳传输延迟 = 5 秒

稍后将有更多关于延迟的内容 ...

# Packet switching versus circuit switching

*packet switching allows more users to use network!*

example:
- 1 Mb/s link
- each user:
  - 100 kb/s when "active"
  - active 10% of time




- *circuit-switching:*
  - 10 users
- *packet switching:*
  - with 35 users, probability > 10 active at same time is less than .0004 *

*Q:* how did we get value 0.0004?

*Q:* what happens if > 35 users ?

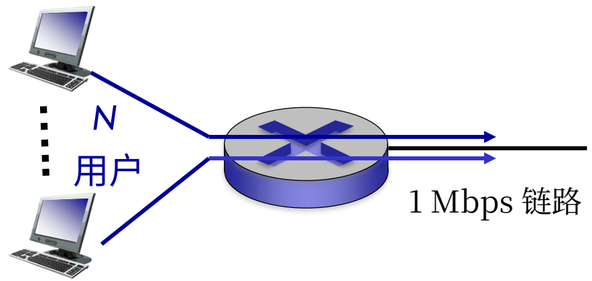
* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive/



# 分组交换与电路交换

分组交换允许更多用户使用网络!

示例:
- 1 Mb/s链路
- 每个用户:
  - 100 kb/s 当"活跃"时
  - 活跃 10% 的时间




- 电路交换:
  - 10 用户
- 分组交换:
  - 有 35 个用户，同时 > 10 活跃的概率小于 .0004 *

*Q:* 我们是如何得到 0.0004 这个值的?

*Q:* 如果 > 35 用户呢?

* 查看 mo 的在线交互练习

更多示例: http://gaia.cs.umass.edu/kurose_ross/interactive /

# Packet switching versus circuit switching

- Probability that a user is active p=0.1
- Probability that a user is inactive q=1-p
- We want to find N such that concurrent users are more than 10

$$\Pr\{Aggregate\ input > 1Mbps\} < 0.001$$

$$\Pr\{k\ out\ of\ \ N\ are\ active\ and\ k > 10\} < 0.001$$

$$\sum_{i=11}^{N} \binom{N}{i} p^i (1-p)^{N-i} < 0.001$$



# 分组交换与电路交换

▪用户活跃概率 p=0.1 ▪ 用户非活跃概率 q=1-p ▪ 我们需要找到 N，使得并发用户数超过 10

> <Mbps 输入聚合 001.0}1 Pr{

$$\sum_{=}^{N} \binom{}{} 1- < \quad i \quad iNi \quad ppiN_{11}$$

<活动Nofoutk 001.0}10 Pr{

0010.)(

# Packet switching versus circuit switching

## is packet switching a "slam dunk winner?"

- great for bursty data
  - resource sharing
  - simpler, no call setup
- excessive congestion possible: packet delay and loss
  - protocols needed for reliable data transfer, congestion control
- Q: How to provide circuit-like behavior?
  - bandwidth guarantees needed for audio/video apps
  - still an unsolved problem (chapter 7)

Q: human analogies of reserved resources (circuit switching) versus on-demand allocation (packet-switching)?



# 分组交换与电路交换

## 分组交换是"稳操胜券"吗?

- 非常适合突发数据
  - 资源共享 • 更简单，无需呼叫建立
- 可能存在过度拥塞：数据包延迟和丢失
  - 需要可靠数据传输协议，拥塞控制
- Q: 如何提供电路交换行为?
  - 音频/视频应用需要带宽保证 • 仍然是一个未解决的问题(第7章)

Q: 预留资源（电路交换）与按需分配（分组交换）的人类类比?

# Internet structure: network of networks

- End systems connect to Internet via access ISPs (**Internet Service Providers**)
  - residential, company and university ISPs
- Access ISPs in turn must be interconnected.
  - so that any two hosts can send packets to each other
- Resulting network of networks is very complex
  - evolution was driven by economics and national policies
- Let's take a stepwise approach to describe current Internet structure

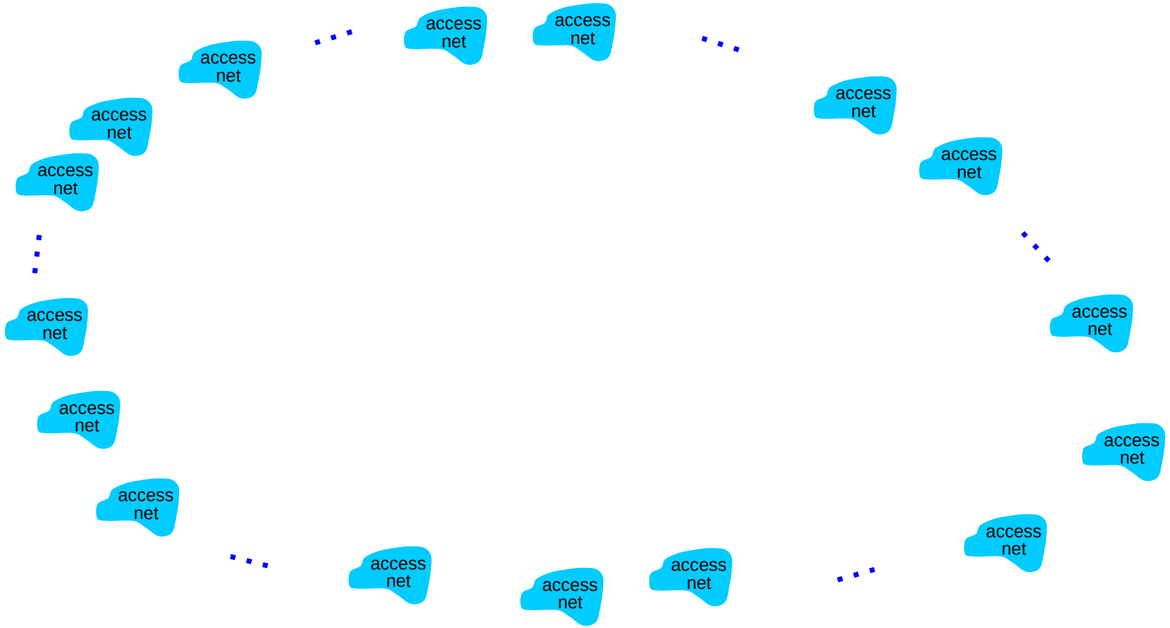


# 互联网结构：网络中的网络

- 终端系统通过 接入互联网服务提供商(**互联网服务提供商**)连接到互联网
  - 住宅、公司和大学互联网服务提供商
- 接入互联网服务提供商必须相互连接。
  - 这样任何两个主机都可以互相发送数据包
- 由此形成的网络中的网络非常复杂
  - 其发展是由经济和国家政策推动的
- 让我们逐步描述当前的互联网结构

# Internet structure: network of networks

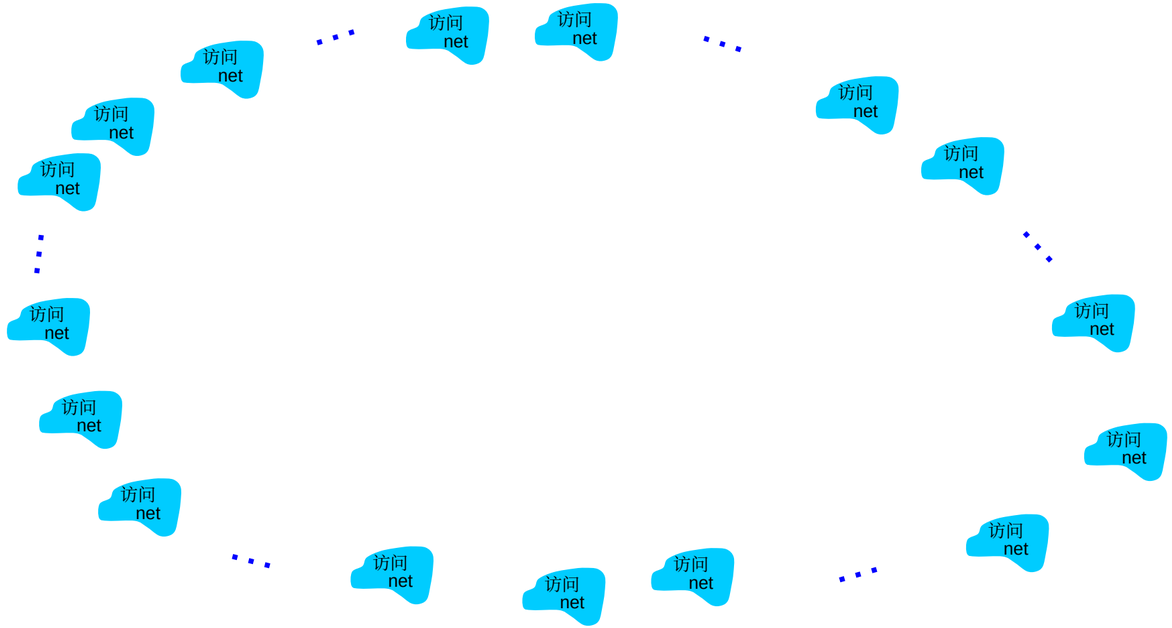
*Question:* given *millions* of access ISPs, how to connect them together?





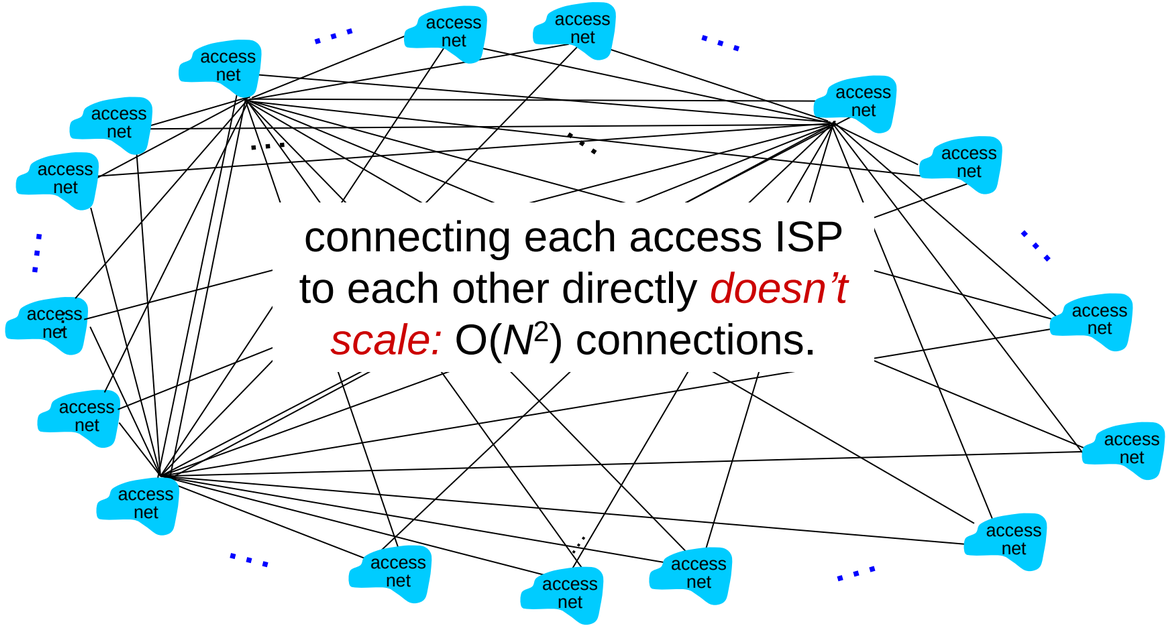

# 互联网结构：网络中的网络

问题： 给定 数百万 互联网服务提供商，如何将它们连接在一起?



# Internet structure: network of networks

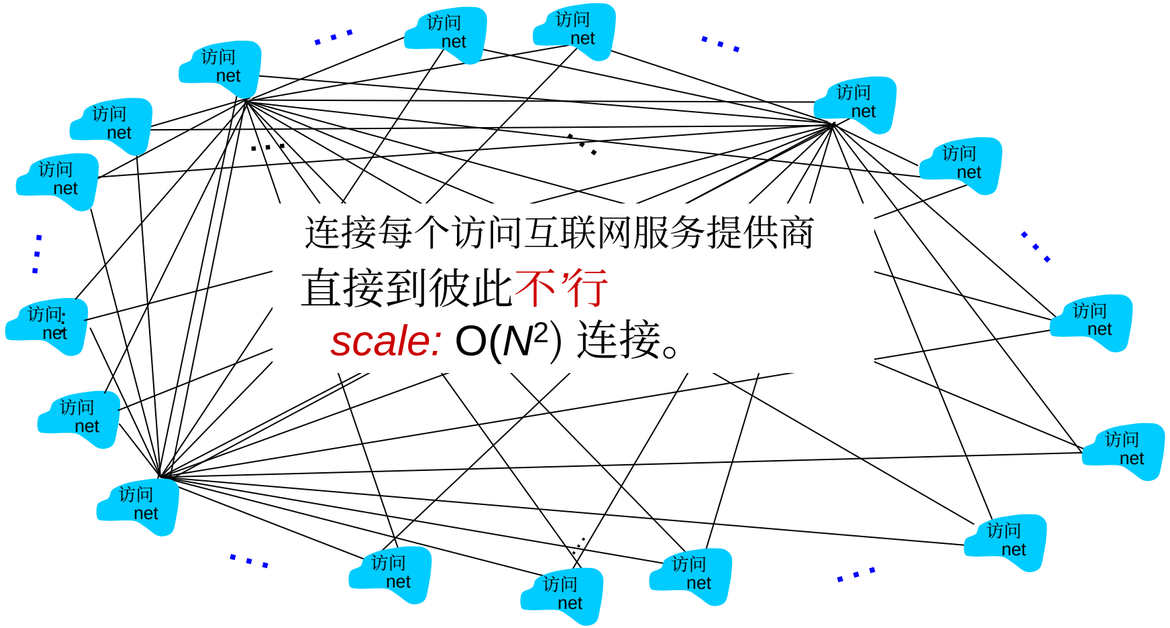
*Option: connect each access ISP to every other access ISP?*





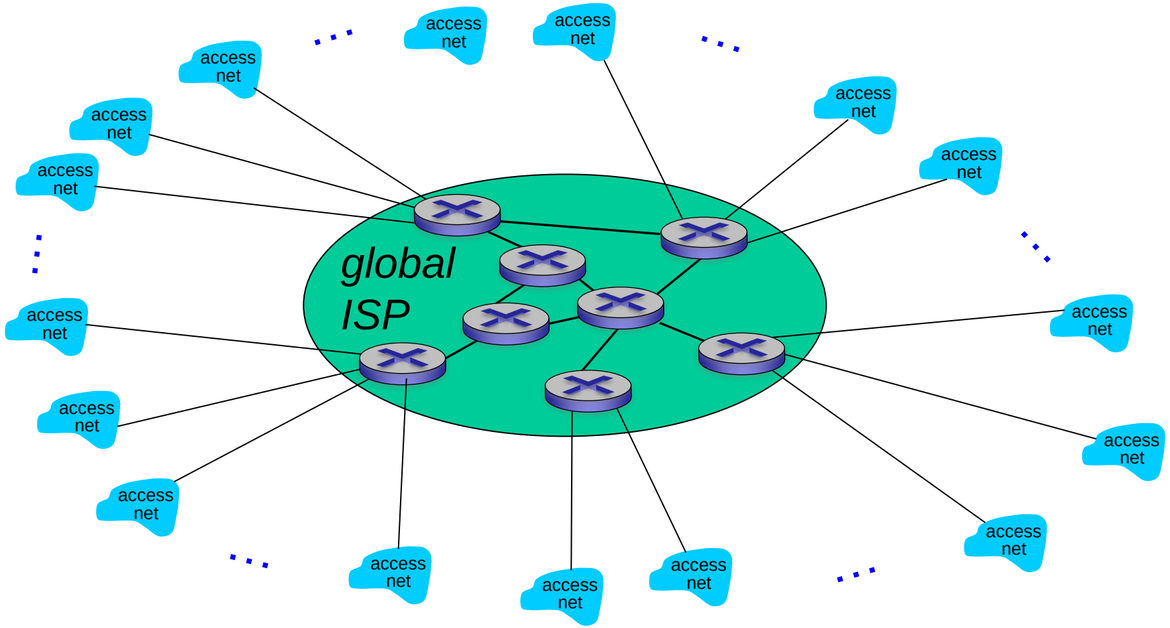

# 互联网结构：网络中的网络

选项：将每个访问互联网服务提供商连接到其他每个访问互联网服务提供商？



# Internet structure: network of networks

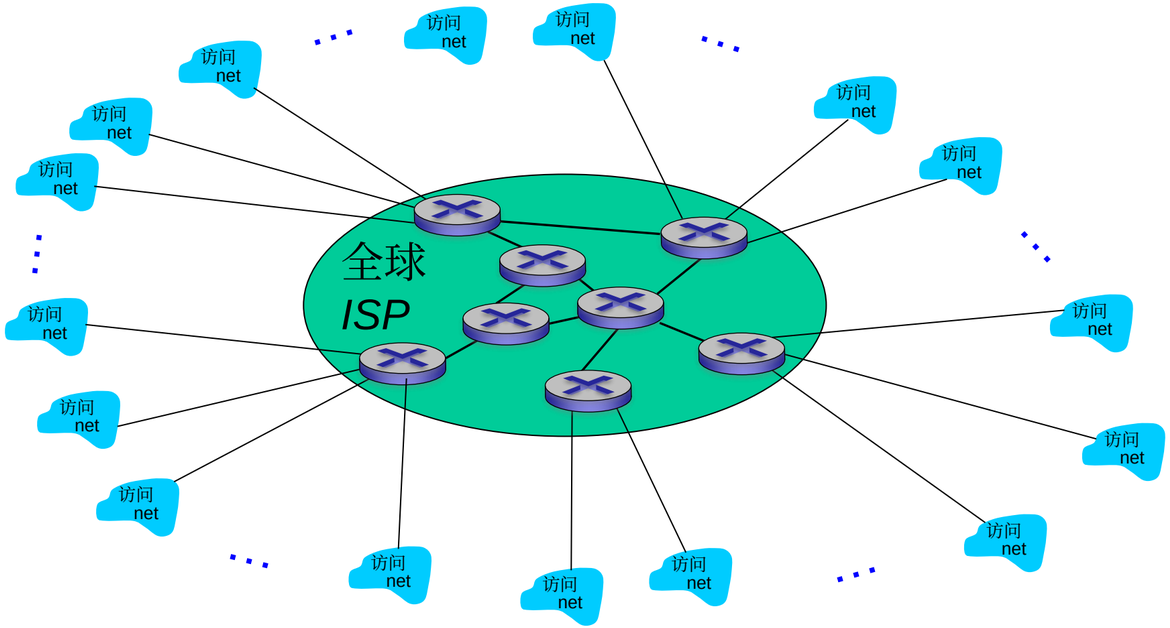
*Option: connect each access ISP to one global transit ISP?*
*Customer and provider ISPs have economic agreement.*





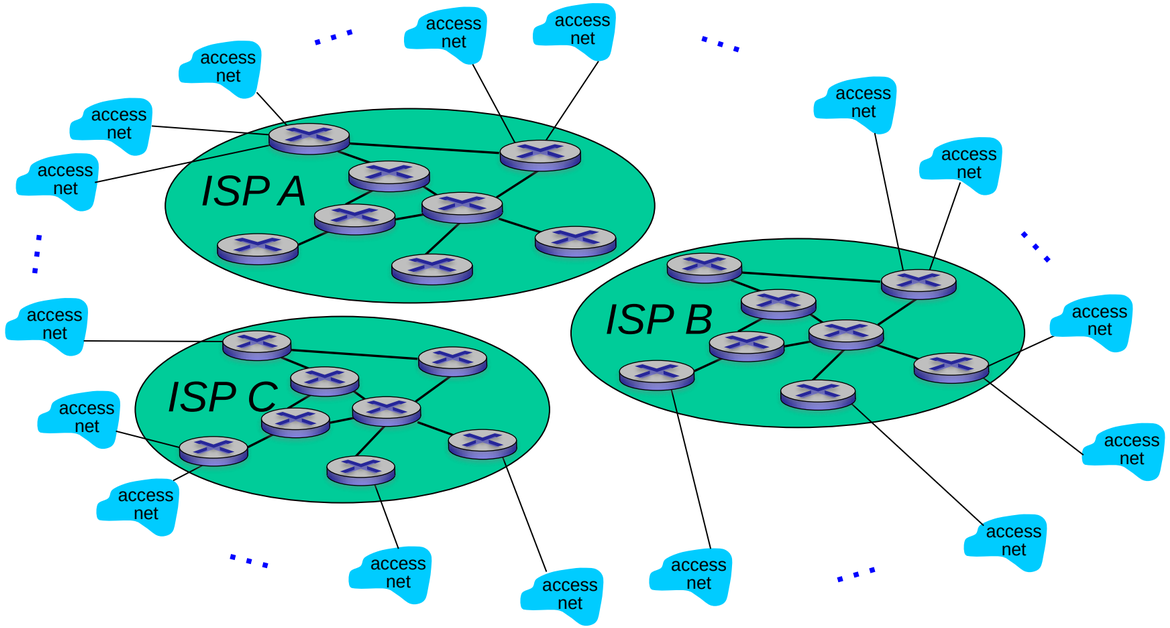

# 互联网结构：网络中的网络

选项： 将每个访问互联网服务提供商连接到一个全球传输互联网服务提供商？客户 和 提供者 互联网服务提供商之间有经济协议。



# Internet structure: network of networks

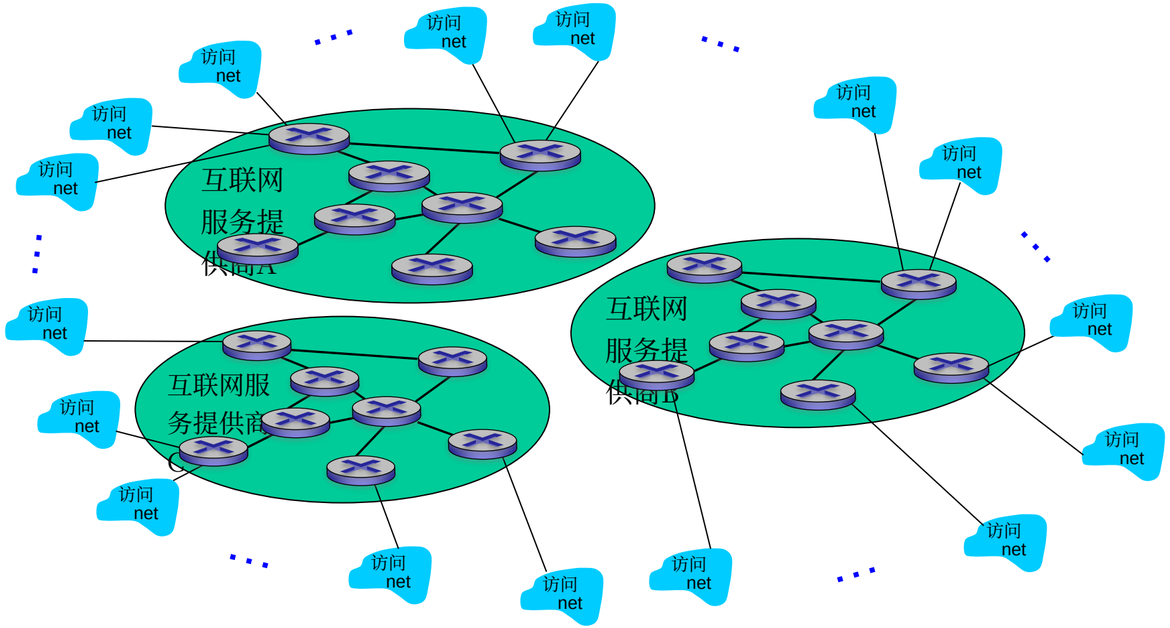
But if one global ISP is viable business, there will be competitors ….





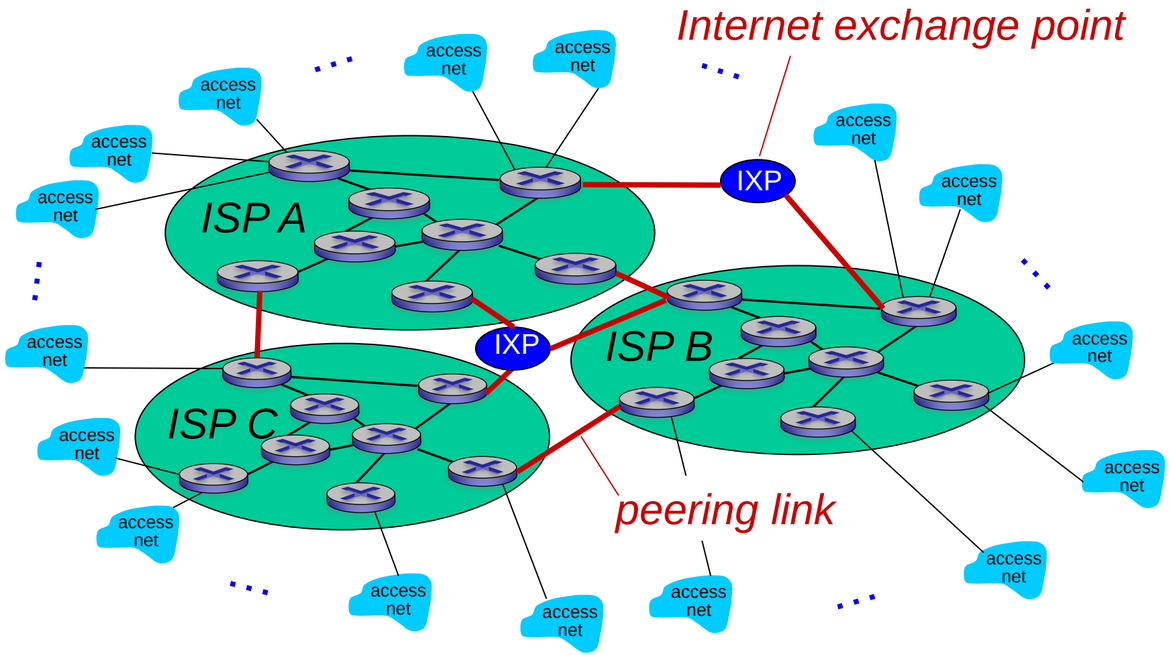

# 互联网结构：网络中的网络

但若一家全球互联网服务提供商的商业模式可行，就会有竞争对手……



# Internet structure: network of networks

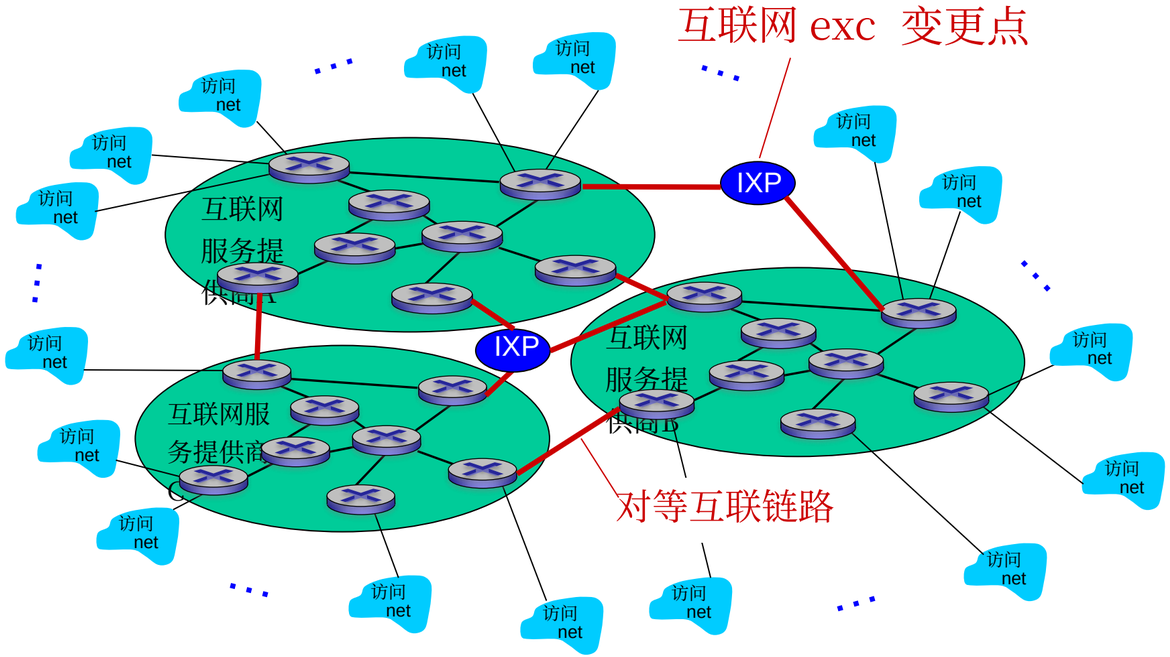
But if one global ISP is viable business, there will be competitors …. which must be interconnected





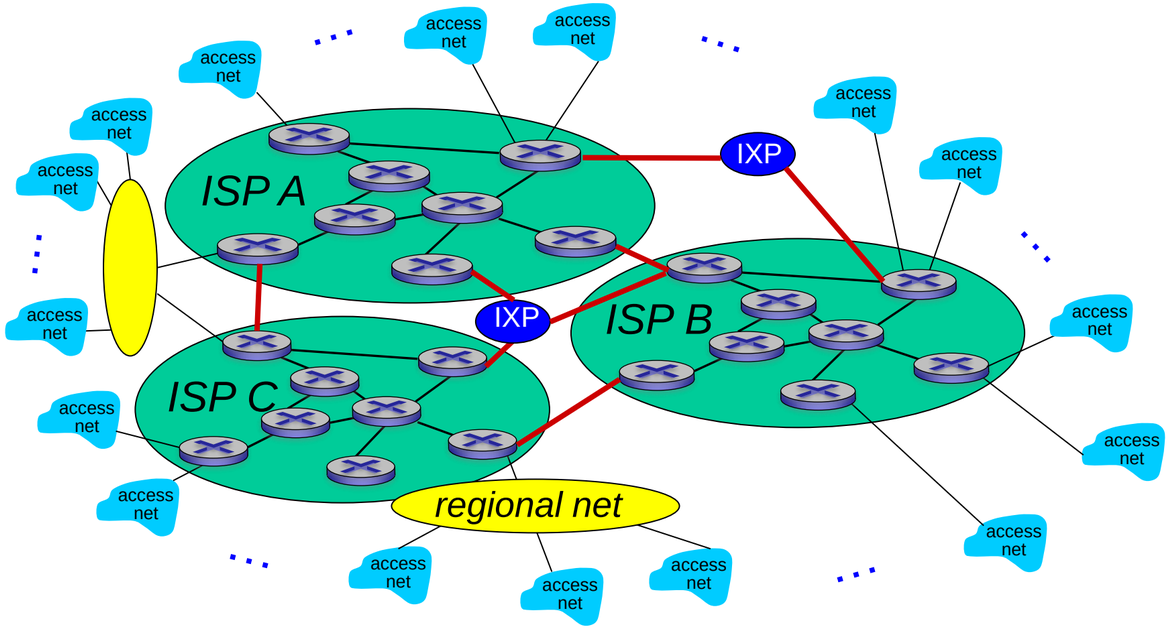

# 互联网结构：网络中的网络

但如果一个全球互联网服务提供商是可行的商业，就会有竞争对手……这些竞争对手必须互联



# Internet structure: network of networks

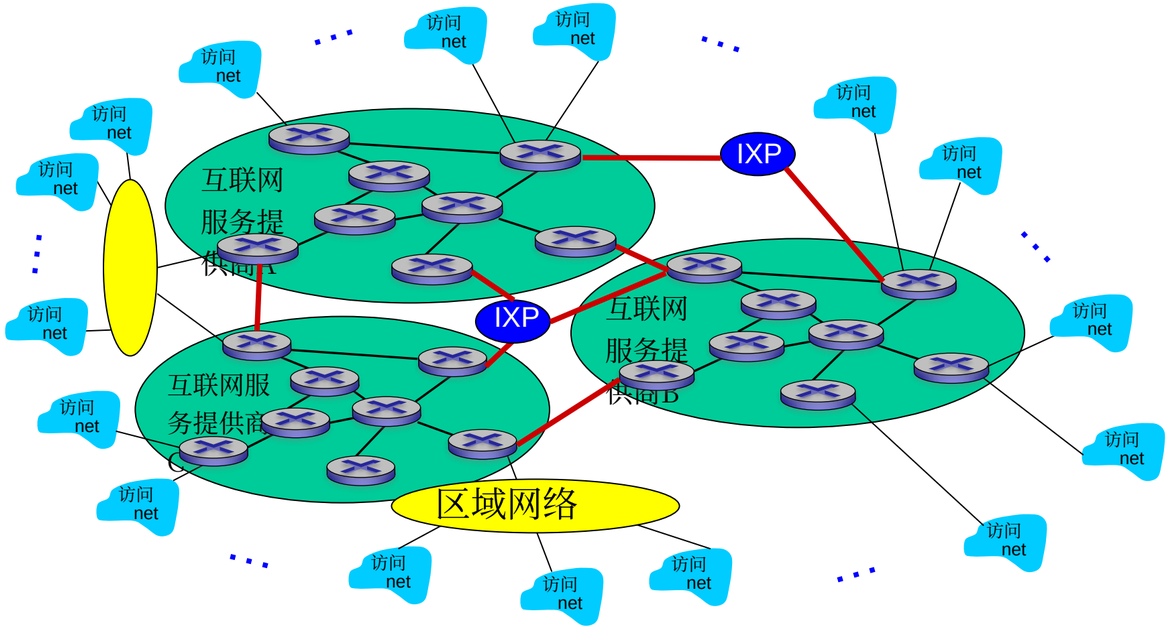
… and regional networks may arise to connect access nets to ISPs





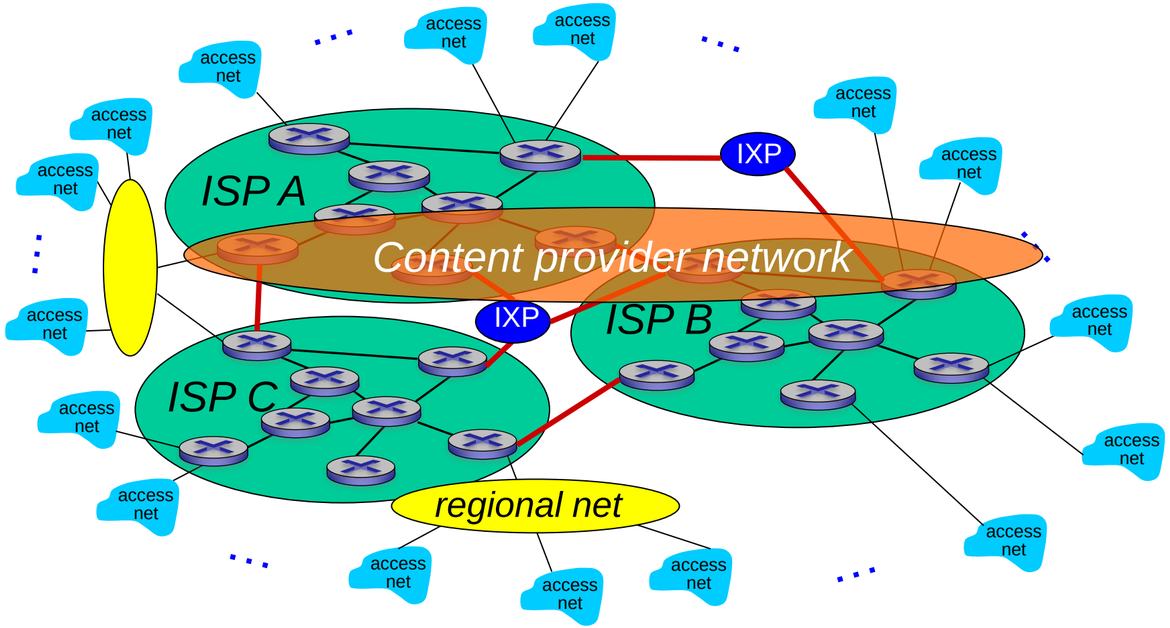

# 互联网结构：网络中的网络

...和区域网络可能会出现，以连接接入网到互联网服务提供商



# Internet structure: network of networks

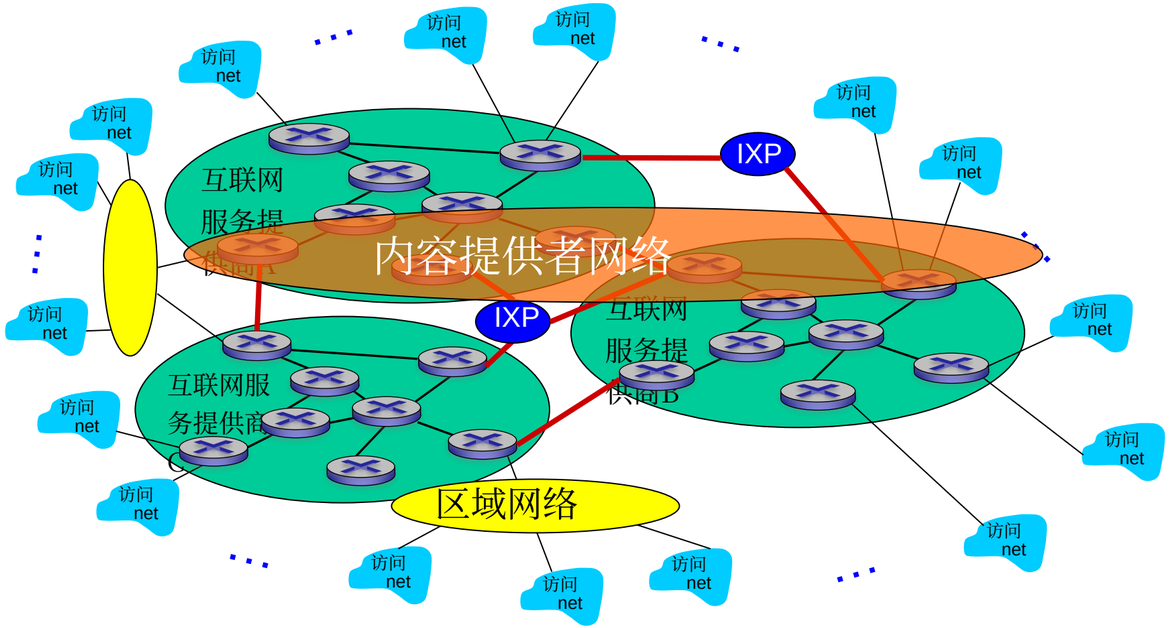
… and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users



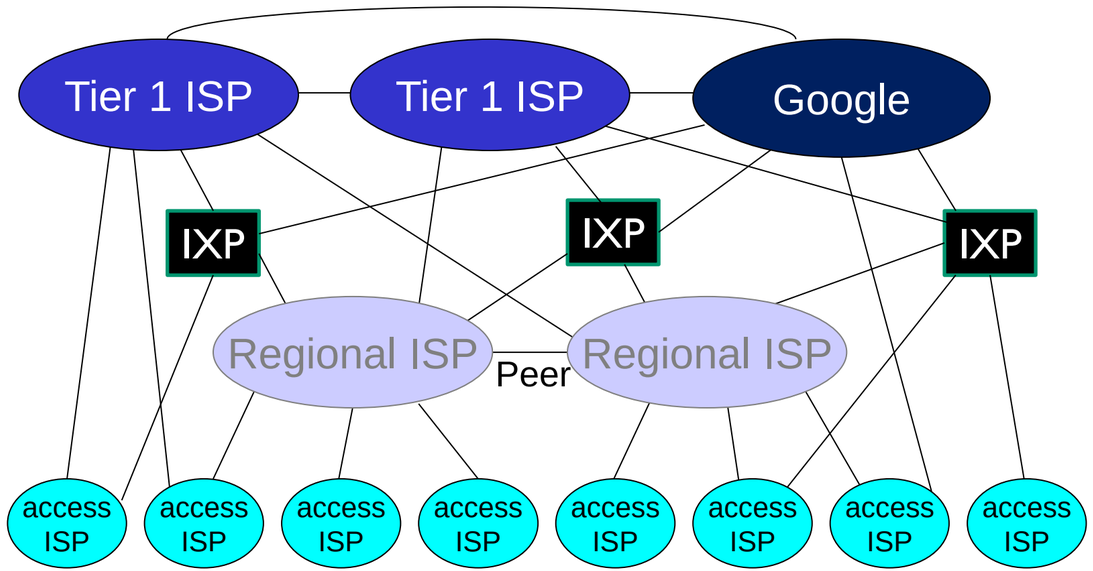



# 互联网结构：网络中的网络

...并且内容提供者网络（例如Google、Microsoft、Akamai）可能会自行运营网络，以将服务、内容靠近最终用户



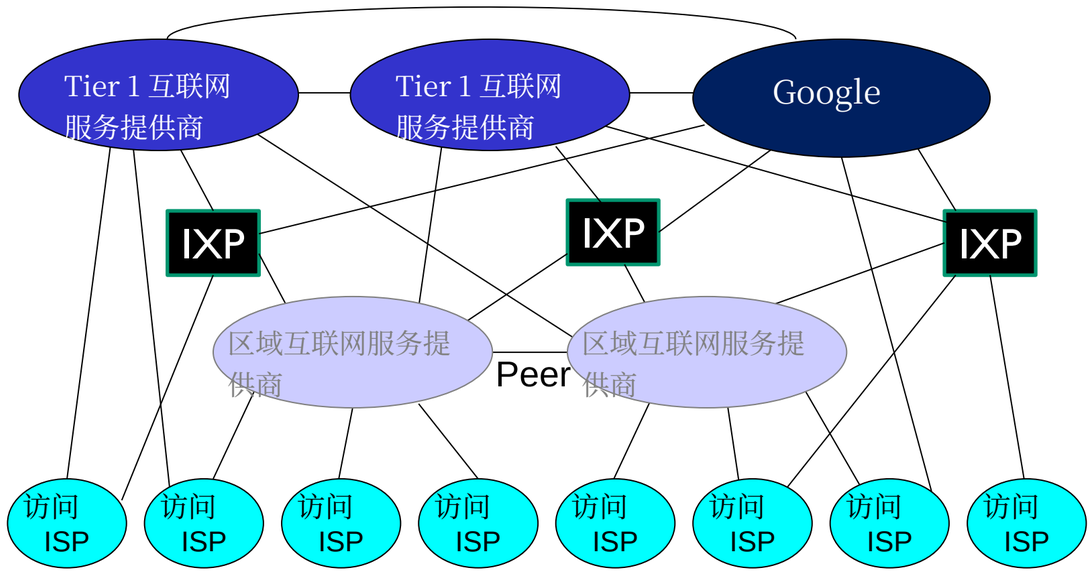
# Internet structure: network of networks




- access ISPs, regional ISPs, tier-1 ISPs,
- PoPs, multi-homing, peering, and IXPs

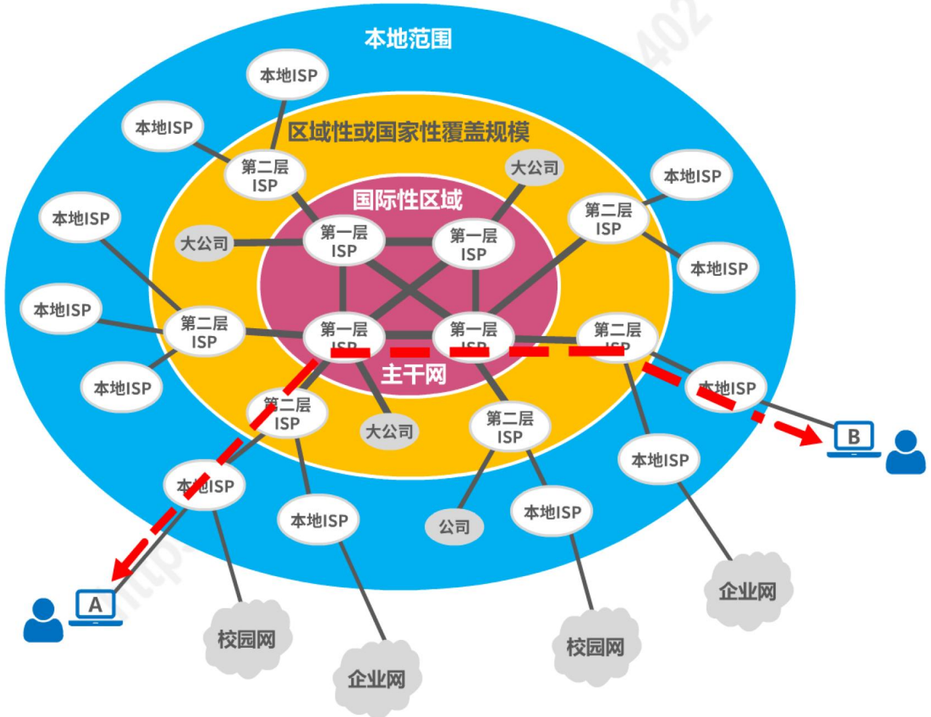


# 互联网结构：网络中的网络



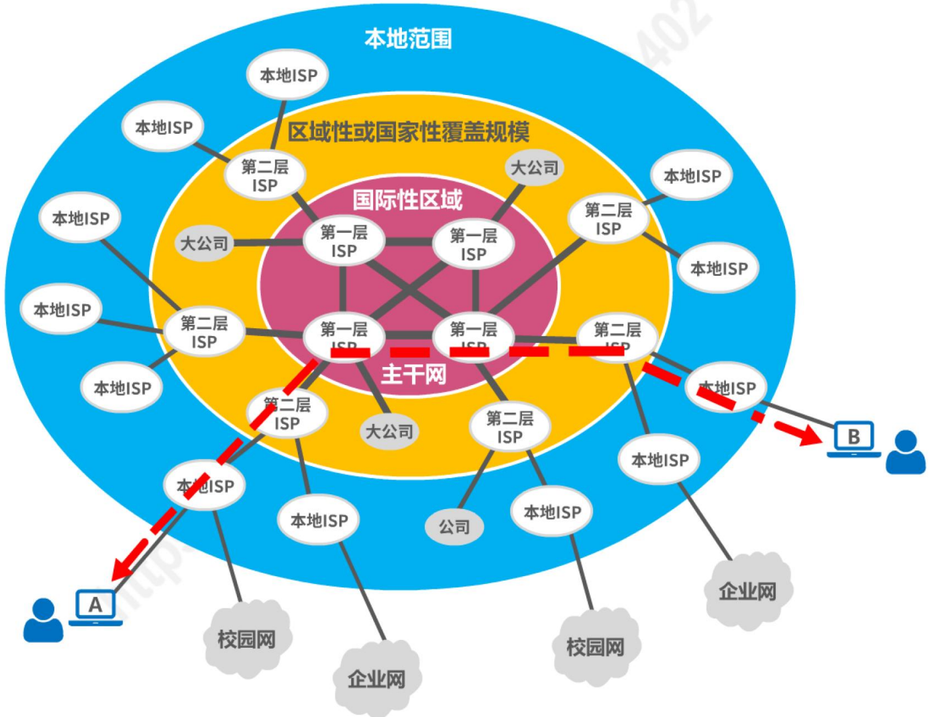

- 访问互联网服务提供商、区域互联网服务提供商、一级互联网服务提供商、▪接入点、多宿主、对等连接和IXP

# Internet structure: network of networks






# 互联网结构：网络中的网络



# Chapter 1: roadmap

1.1 What is the Internet?
1.2 Network edge
- hosts, access network, physical media

1.3 Network core
- packet switching, circuit switching, network structure

1.4 Delay & loss in packet-switched networks
1.5 Protocol layers, Service models
1.6* Internet history



# 第一章：路线图

1.1 什么是互联网?
1.2 网络边缘
- 主机、接入网、物理媒体

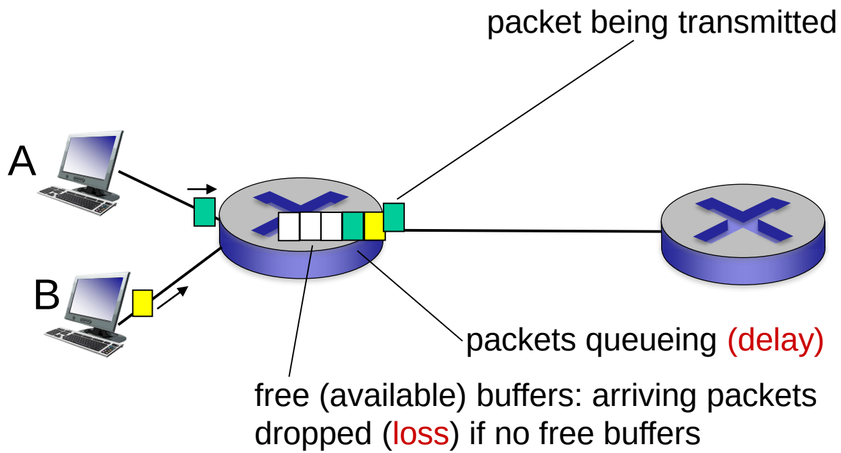
1.3 网络核心
- 数据包交换、电路交换、网络结构

1.4 数据包交换网络中的延迟与丢失
1.5 协议层、服务模型
1.6* 互联网历史

# How do loss and delay occur?

packets *queue* in router buffers

- packet arrival rate to link (temporarily) exceeds output link capacity
- packets queue, wait for turn

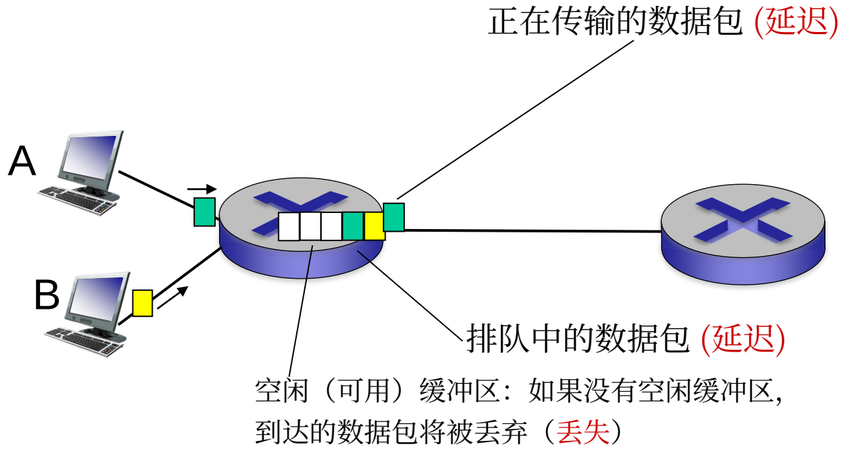




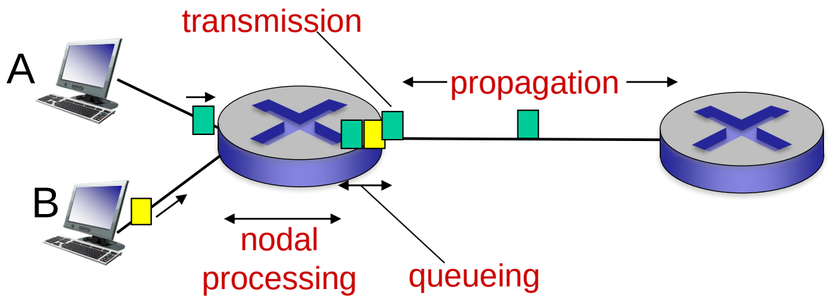

# 丢失和延迟是如何发生的?

数据包 排队在路由器缓冲区

- 数据包到达速率（暂时）超过输出链路容量
- 数据包排队，等待轮次



# Four sources of packet delay




$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

$d_{proc}$: nodal processing

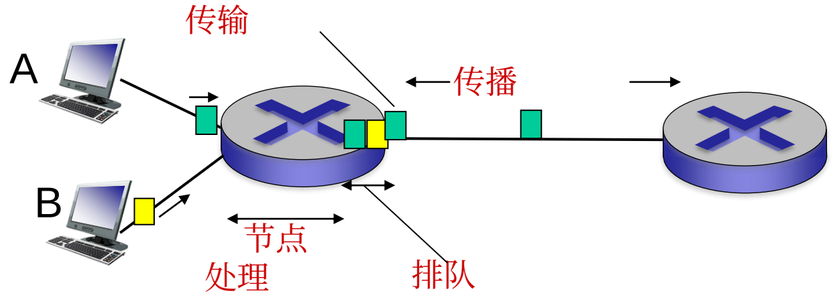
- check bit errors
- determine output link
- typically < msec

$d_{queue}$: queueing delay

- time waiting at output link for transmission
- depends on congestion level of router



# 数据包延迟的四个来源




$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

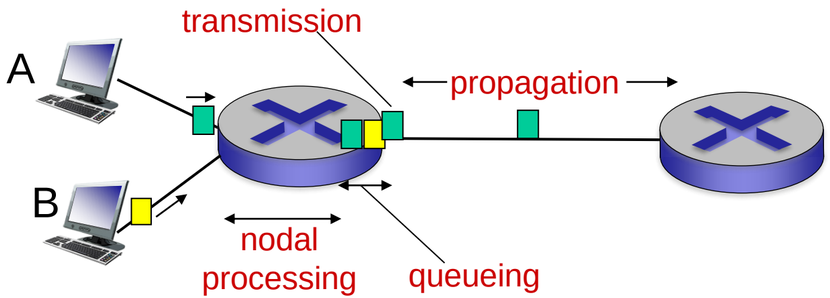
$d^{proc}$：节点处理

- 检查比特错误
- 确定输出链路
- 通常 < 毫秒

$d_{queue}$: 排队延迟

- 等待在输出链路传输的时间
- 取决于路由器的拥塞程度

# Four sources of packet delay




$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

$d_{trans}$: transmission delay:
- $L$: packet length (bits)
- $R$: link *bandwidth (bps)*
- $d_{trans} = L/R$

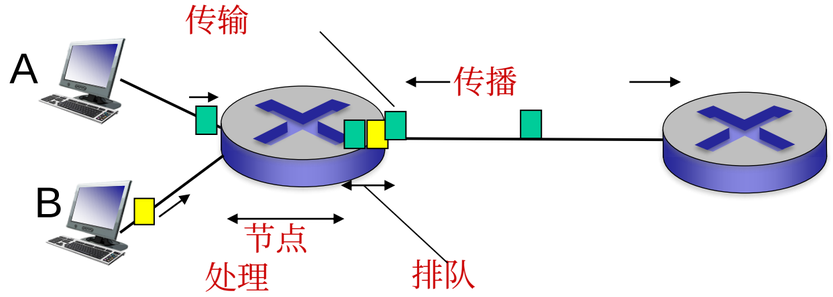
$d_{trans}$ and $d_{prop}$ *very* different

$d_{prop}$: propagation delay:
- $d$: length of physical link
- $s$: propagation speed (~2x10$^8$ m/sec)
- $d_{prop} = d/s$

* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive/
* Check out the Java applet for an interactive animation on trans vs. prop delay



# 数据包延迟的四个来源




$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

$d_{trans}$: 传输延迟:
- $L$: 数据包长度 (比特)
- $R$: 链路 带宽 (比特/秒)
- $d_{传输} = L/R$

$d_{trans}$ 和 $d_{prop}$ 非常不同

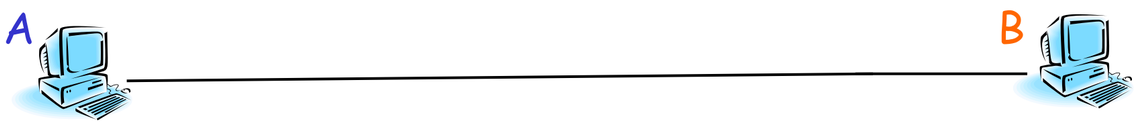
$d_{prop}$: 传播延迟:
- $d$: 物理链路长度
- $s$: 传播速度 (~2x10$_8$m/sec)
- $d_{prop} = d/s$

* 查看在线交互练习获取更多示例：http://gaia.cs.umass.edu/kurose_ross/interactive/
* 查看Java小程序，了解传输延迟与传播延迟的交互式动画

# Excersice-1(nodal delay)

Assuming that the link length *d* between two hosts is 60m, the link bandwidth **R** is 10Mbps, and the signal propagation rate *s* is $2x10^8$ m / s. Now host A sends 1-bit packet to host B.

1) How long does it take for host B to receive this packet? If host A starts transmitting the packet at time *t = 0*, how far does the front end of the signal of the packet spread at time *t =* $d_{trans}$?
2) What if the link length *d* is 20 meters?
3) If the link length *d* is 10 meters, what is the specific transmission situation?
4) If host A sends n-bit packet to host B, how long does it take for host B to receive the packet?





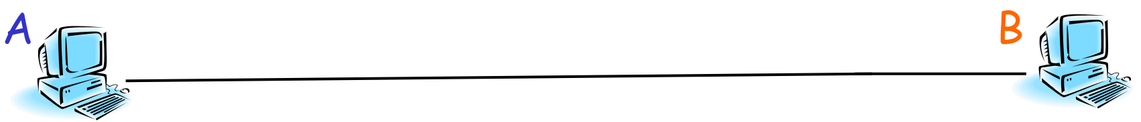

# 练习-1(节点延迟)

假设两台主机之间的链路长度 *d* 为60米，链路带宽 **R**为10Mbps，信号传播速度 *s* 为$2x10^8$ 米/秒。现在主机A向主机B发送 1比特 的数据包。

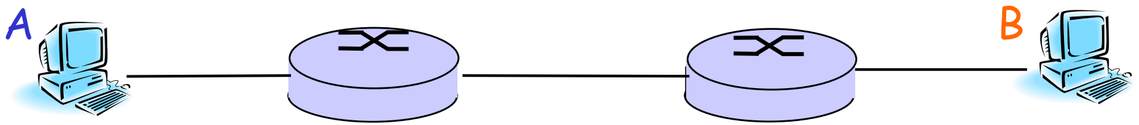
1) 主机B接收这个数据包需要多长时间？如果主机A在时间 *t = 0*开始传输数据包，那么在时间 *t =* 数据包的信号前沿传播多远？

2) 如果链路长度 *d* 是20米呢？
3) 如果链路长度 *d* 是10米，具体的传输情况是什么？

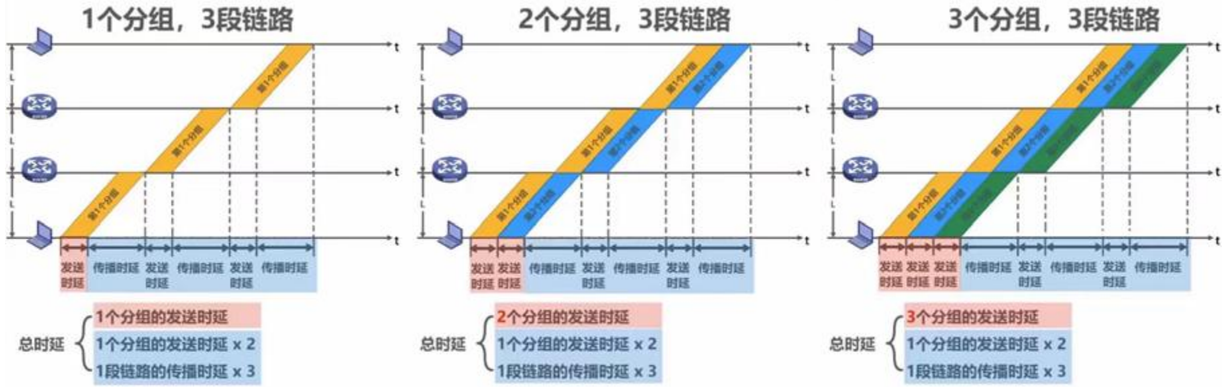
4) 如果主机A向主机B发送n位数据包，主机B接收数据包需要多长时间？



# End-to-end delay



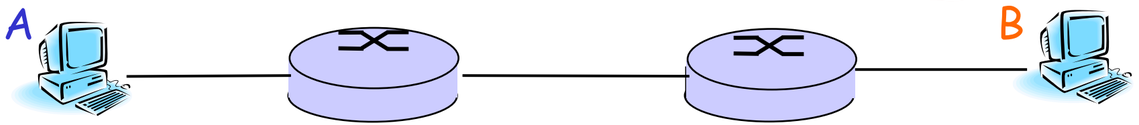

假设：分组等长，各链路长度相同、带宽也相同，忽略路由器的处理时延



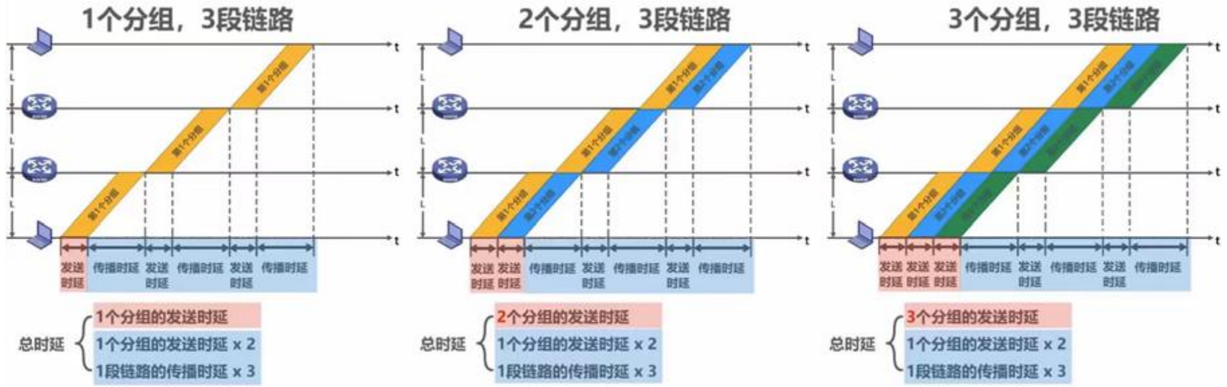

若n个分组，m段链路，则总时延是多少?



# 端到端延迟

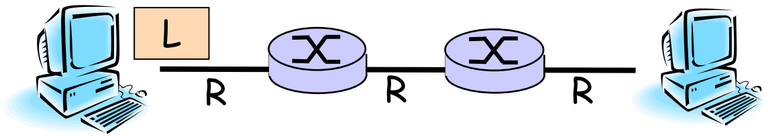



假设：分组等长，各链路长度相同、带宽也相同，忽略路由器的处理时延




若n个分组，m段链路，则总时延是多少?

# Exercise-2 (end-to-end delay)



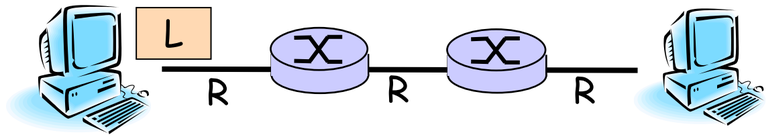

- Message switching and Packet switching: Entire message or packet must arrive at router before it can be transmitted on next link: *store and forward*
- Calculate the end-to-end delay of message switching and packet switching

Example:

- L = 7.5 Mbits
- R = 1.5 Mbps
- $t_{prop}$ : propagation delay of each link
- Divide the message into 5000 small packets: l=1.5 kbits



# 练习-2 (端到端延迟)




- 消息交换和分组交换：整个消息或分组必须到达路由器才能在下一个链路上传输：存储转发

- 计算消息交换和分组交换的端到端延迟

示例：

- 数据包长度 = 7.5 Mbits
- 链路带宽 = 1.5 Mbps
- tprop：每个链路的传播延迟
- 将消息分成 5000 个小数据包：l=1.5kbits

# Exercise-3 (end-to-end delay)

The message to be sent is *x* (bits) in total. There are *k* links from the source to the destination. The propagation delay of each link is *d*(s) and the bandwidth is *R*(bps). In circuit switching, the circuit establishment time is *s*(s). In packet switching, the message can be divided into several packets with *p*(bits) length. It is assumed that the processing time and queuing time of the message/packets at each node are ignored.

Q: under what conditions is the end-to-end delay of circuit switching greater than that of packet switching?



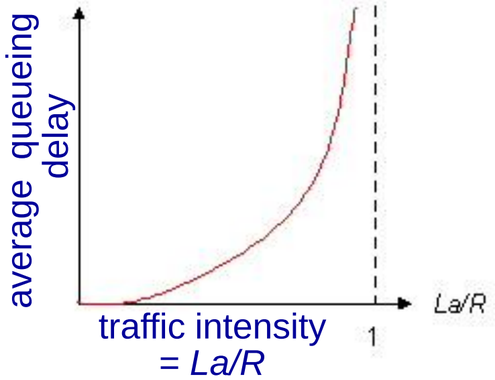
# 练习-3 (端到端延迟)

要发送的消息是*x* (比特) 总计。有*k* 条链路从源端到目的端。每条链路的传播延迟是*d*(秒) ，带宽是*R*(比特每秒)。在电路交换中，电路建立时间是*s*(秒)。在分组交换中，消息可以分成多个数据包，其*p*(比特) 长度。假设每个节点处消息/数据包的处理时间和排队时间都被忽略。

Q：在什么情况下，电路交换的端到端延迟大于分组交换?

# Queueing delay (revisited)

- *R*: link bandwidth (bps)
- *L*: packet length (bits)
- a: average packet arrival rate




- *La/R* ~ 0: avg. queueing delay small
- *La/R* -> 1: avg. queueing delay large
- *La/R* > 1: more "work" arriving than can be serviced, average delay infinite!

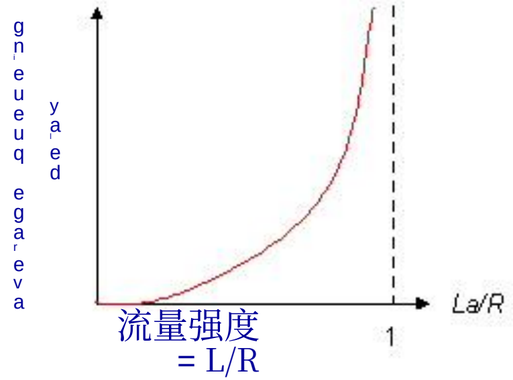



* Check online interactive animation on queuing and loss



# 排队延迟（重访）

- R：链路带宽 (bps) ▪ L：分组长度 (比特) ▪ a：平均分组到达率



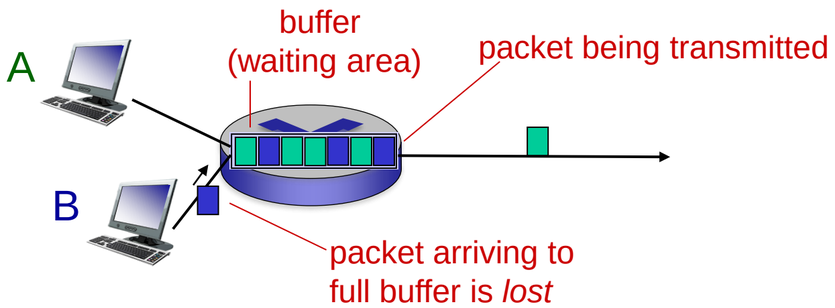

- L/R ~ 0: 平均排队延迟小
- L/R -> 1: 平均排队延迟大
- *La/R* > 1：更多"工作"到达而无法被服务，平均延迟无限！




* 在线检查排队和丢失的交互式动画

# Packet loss

- queue (aka buffer) preceding link in buffer has finite capacity
- when packet arrives to full queue, packet is dropped (aka lost)
- lost packet may be retransmitted by previous node, by source end system, or not at all

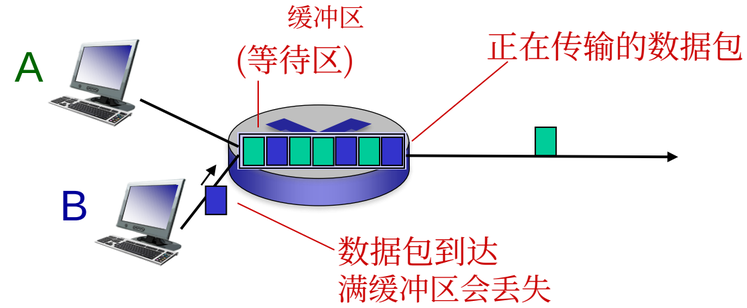



* Check out the Java applet for an interactive animation on queuing and loss



# 数据包丢失

- 队列（又名缓冲区）在链路前面的缓冲区具有有限容量
- 当数据包到达满队列时，数据包会被丢弃（即丢失）
- 丢失的数据包可能由前一个节点、源端系统重新传输，也可能根本不重新传输




* 查看Java小程序，了解排队和丢失的交互式动画

# Chapter 1: roadmap

1.1 What is the Internet?
1.2 Network edge
- hosts, access network, physical media

1.3 Network core
- packet switching, circuit switching, network structure

1.4 Delay & loss in packet-switched networks
1.5 Protocol layers, Service models
1.6* Internet history



# 第一章：路线图

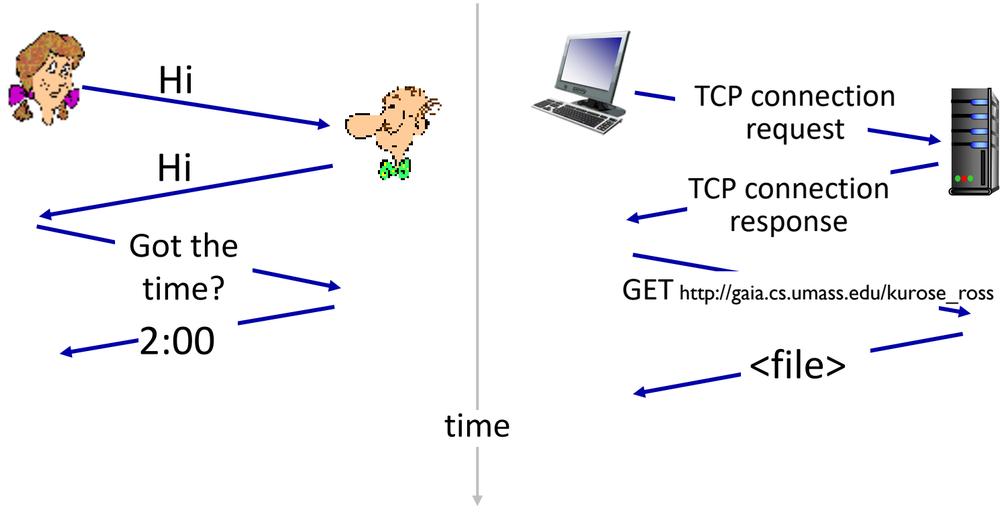
1.1 什么是互联网?
1.2 网络边缘
- 主机、接入网、物理介质

1.3 网络核心
- 分组交换、电路交换、网络结构

1.4 延迟与丢失 - 包交换网络
1.5 协议层与服务模型
1.6* 互联网历史

# What's a protocol?

A human protocol and a computer network protocol:

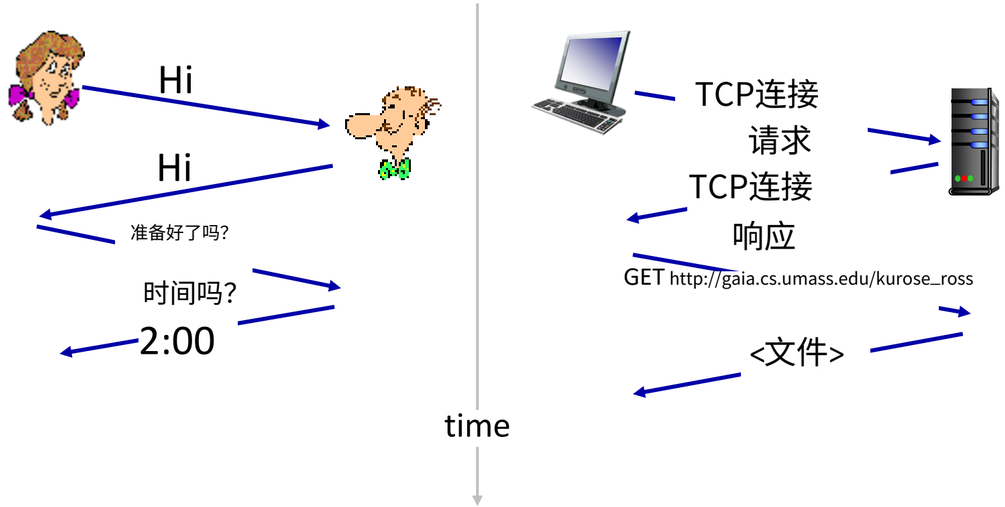



*Q:* other human protocols?



# 什么是协议?

人类协议和计算机网络协议：




*Q:* 其他人类协议?

# What's a protocol?

*Human protocols:*

- "what's the time?"
- "I have a question"
- introductions

*Network protocols:*

- computers (devices) rather than humans
- all communication activity in Internet governed by protocols

Rules for:

… specific messages sent

… specific actions taken when message received, or other events

*Protocols define the format, order of messages sent and received among network entities, and actions taken on message transmission, receipt*



# 什么是协议?

人类协议:

- "现在几点了?"
- "我有问题"
- 介绍

网络协议:

- 计算机(设备)而不是人类
- 互联网中所有通信活动均由协议管理

规则:

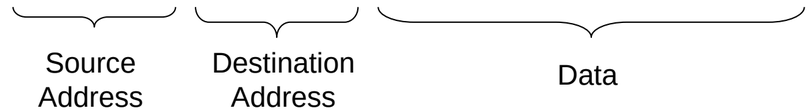
...发送特定消息...当消息被接收时采取特定操作,或其他事件

协议定义了消息的格式、顺序以及网络实体之间发送和接收消息的操作,以及对消息传输和接收

# 3 elements of a protocol

- Syntax: Defines how to do it

Example: 101011011011101101001100111 0...




- Semantics: Defines what it is
- Timing: Defines the order and speed



# 协议的3个要素

- 语法：定义如何执行

示例：101011011011101101001100111 0...

源地址 目的地址 数据

- 语义：定义了它是什么
- 时序：定义了顺序和速率

# Protocol "layers"

*Networks are complex, with many "pieces":*

- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

*Question:*

is there any hope of *organizing* structure of network?

…. or at least our discussion of networks?



# 协议"分层"

网络很复杂，包含许多"部分"：

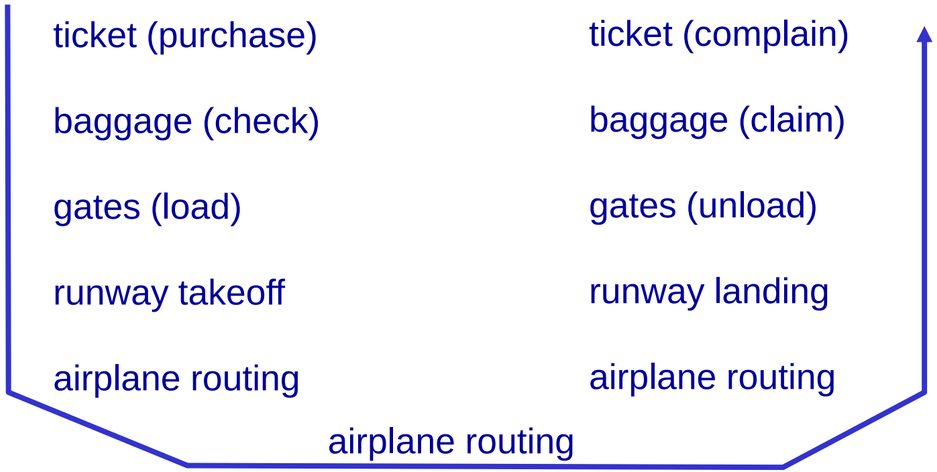
- 主机
- 路由器
- 各种介质的链路
- 应用
- 协议
- 硬件，软件

问题：

是否有希望组织网络结构?

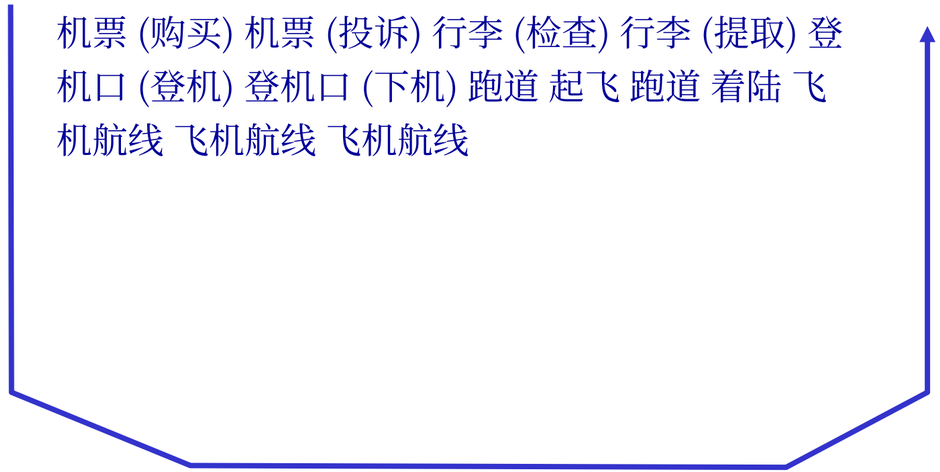
……或者至少是我们关于网络的讨论?

# Organization of air travel



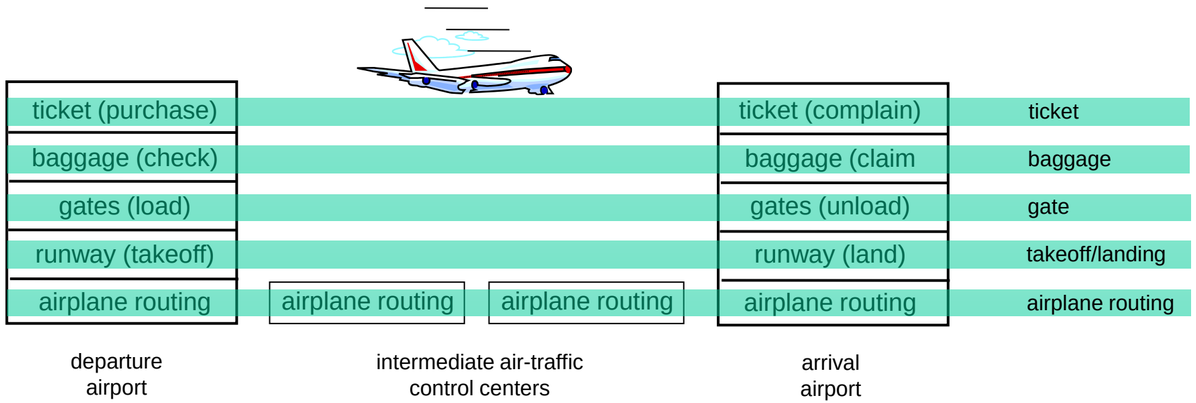

- a series of steps

# 航空旅行组织




- 一系列步骤

# Layering of airline functionality

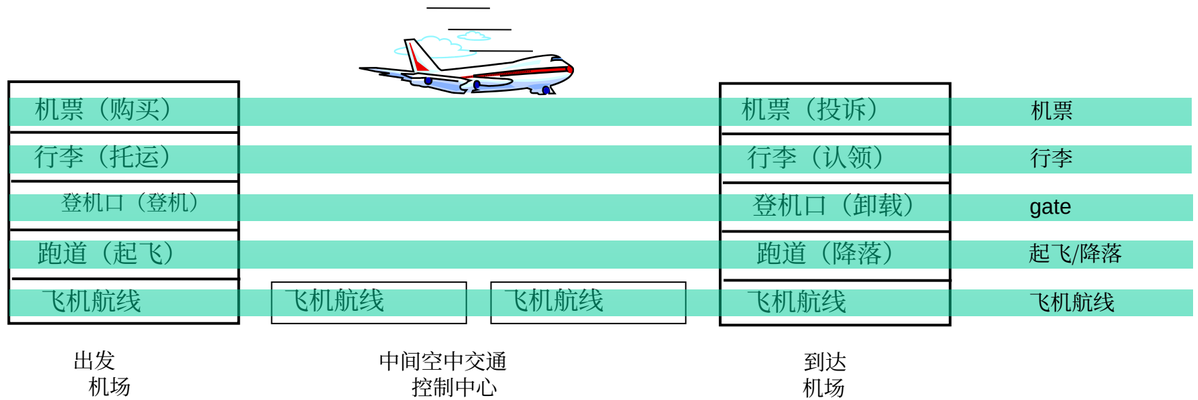



*layers:* each layer implements a service

- via its own internal-layer actions
- relying on services provided by layer below



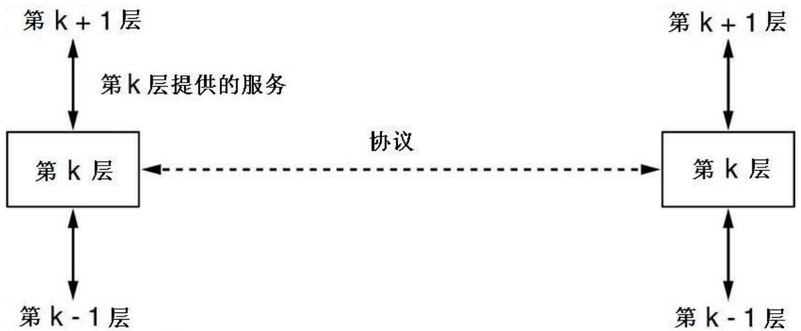
# 航空公司功能分层




分层： 每一层实现一项服务

- 通过其自身的内部层操作
- 依赖于下层提供的服务

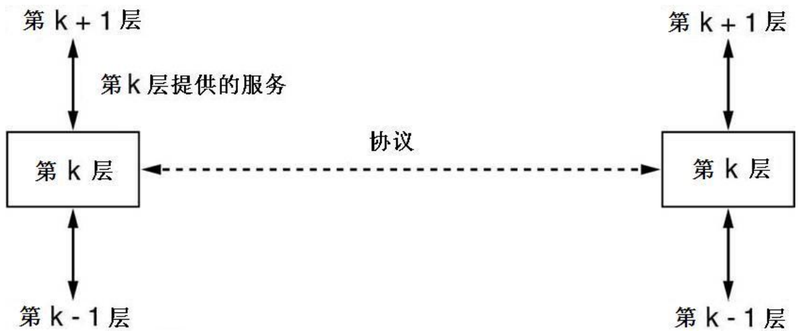
# Layering of airline functionality




*layers:* each layer implements a service
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# 航空公司功能的分层




层： 每一层实现一个服务
- 通过其自身的内部层操作
- 依赖于下层提供的服务

# Why layering?

dealing with complex systems:

- explicit structure allows identification, relationship of complex system's pieces
  - layered *reference model* for discussion
- modularization eases maintenance, updating of system
  - change of implementation of layer's service transparent to rest of system
  - e.g., change in gate procedure doesn't affect rest of system
- layering considered harmful?

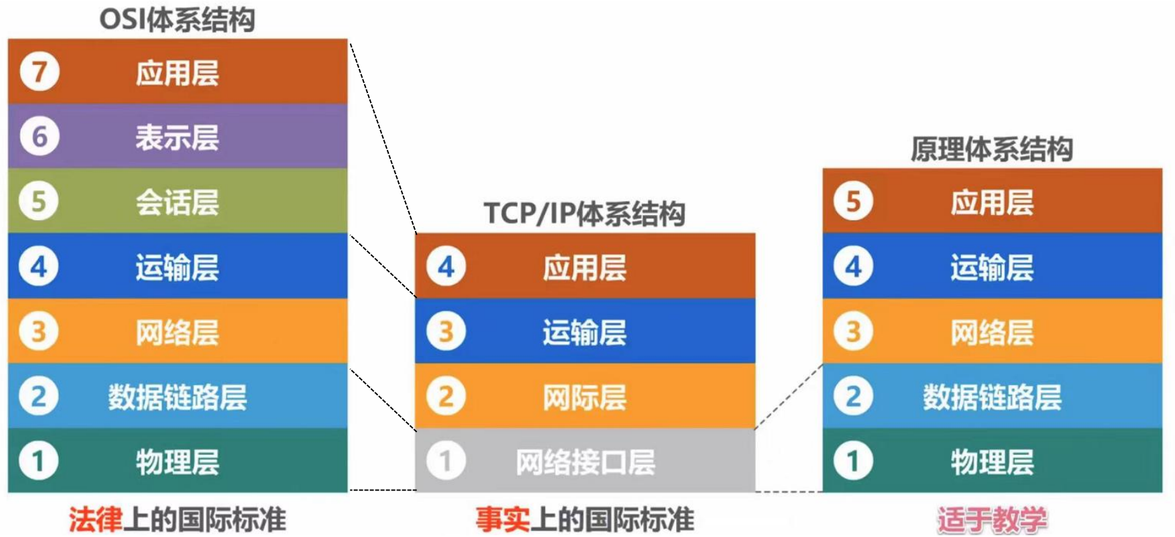


# 为什么分层?

处理复杂系统:

- 显式结构允许识别复杂系统的各个部分及其关系
  - 分层参考模型供讨论
- 模块化简化了系统的维护和更新
  - 层' 服务实现的变更对系统其他部分透明
  - 例如，门程序的变化不会影响系统的其余部分
- 分层被认为有害?

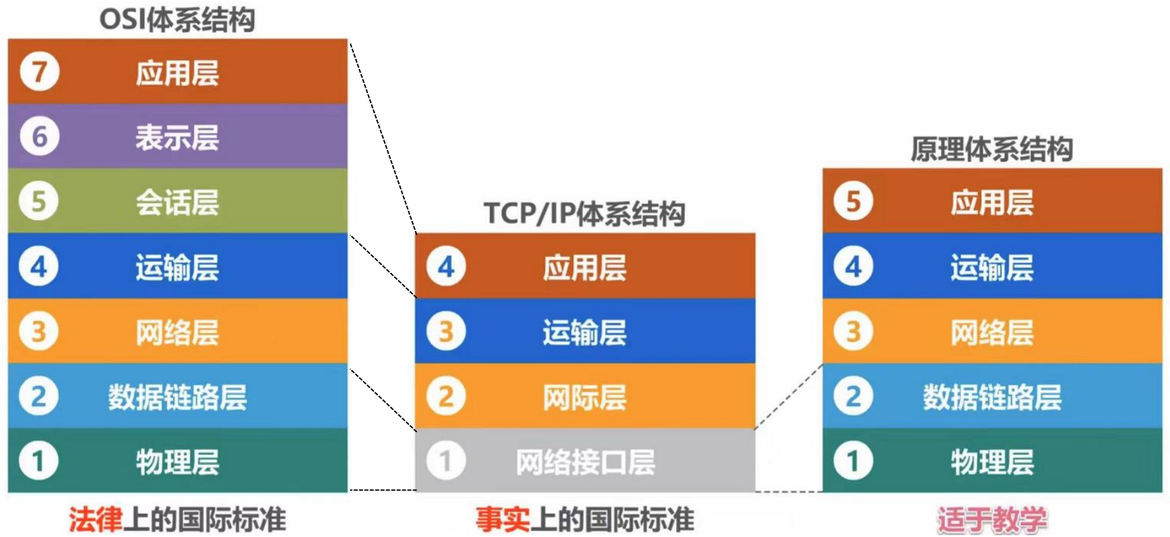
# Network architecture




OSI/RM Internet RM



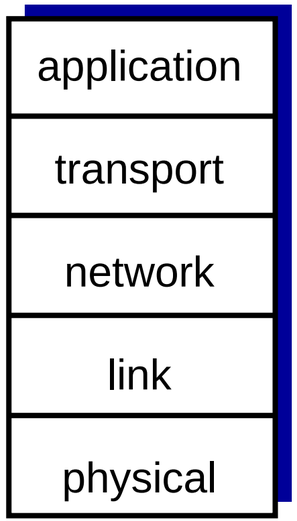
# 网络架构




OSI/RM 互联网RM

# Internet protocol stack

- *application:* supporting network applications
  - FTP, SMTP, HTTP
- *transport:* process-process data transfer
  - TCP, UDP
- *network:* routing of datagrams from source to destination
  - IP, routing protocols
- *link:* data transfer between neighboring network elements
  - Ethernet, 802.111 (WiFi), PPP
- *physical:* bits "on the wire"



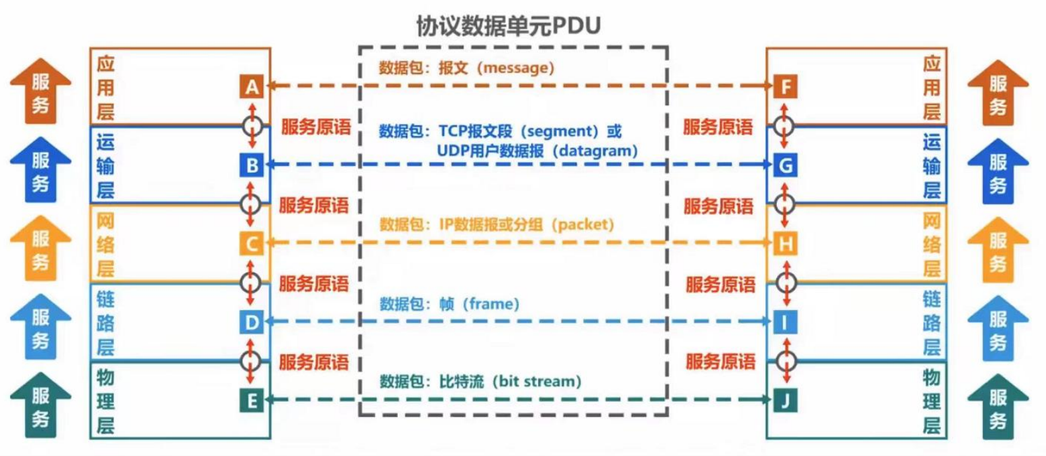



# 互联网协议栈

- 应用：支持网络应用• FTP、SMTP、HTTP
- 传输：进程间数据传输
  - TCP、UDP
- 网络：数据报从源到目的地的路由
  - IP，路由协议
- 链路：相邻网络元素之间的数据传输
  - 以太网，802.111（WiFi），PPP
- 物理：比特"线路上"



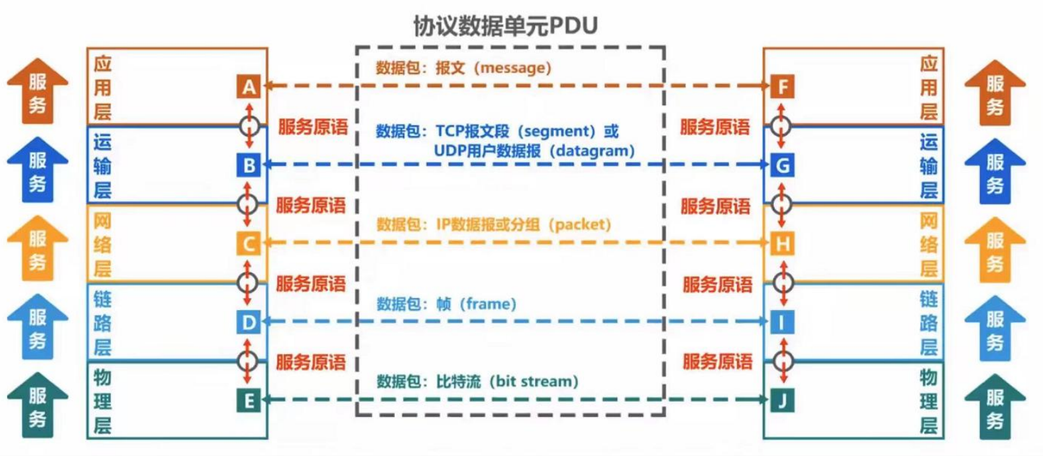
# Terminology of network architecture




**Entity**: hardware and software process that can send and receive information.
**Service**: the function provided by the lower layer to the upper layer
**Service access point (SAP):** places where entities of two adjacent layers interact
**Service primitives**: commands to be exchanged during the interaction between adjacent layers



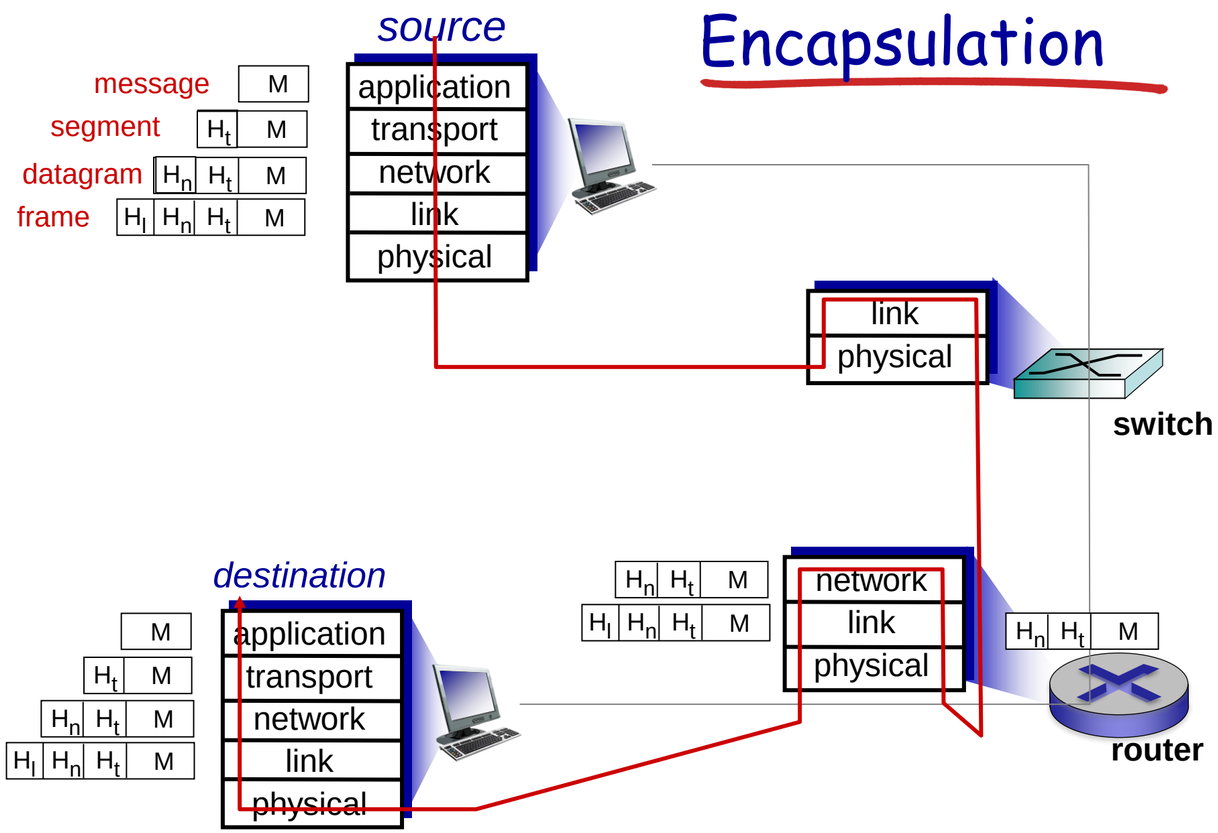
# 网络架构术语




实体：可以发送和接收信息的硬件和软件过程。服务：下层向上层提供的功能服务访问点（SAP）：相邻两层实体交互的位置

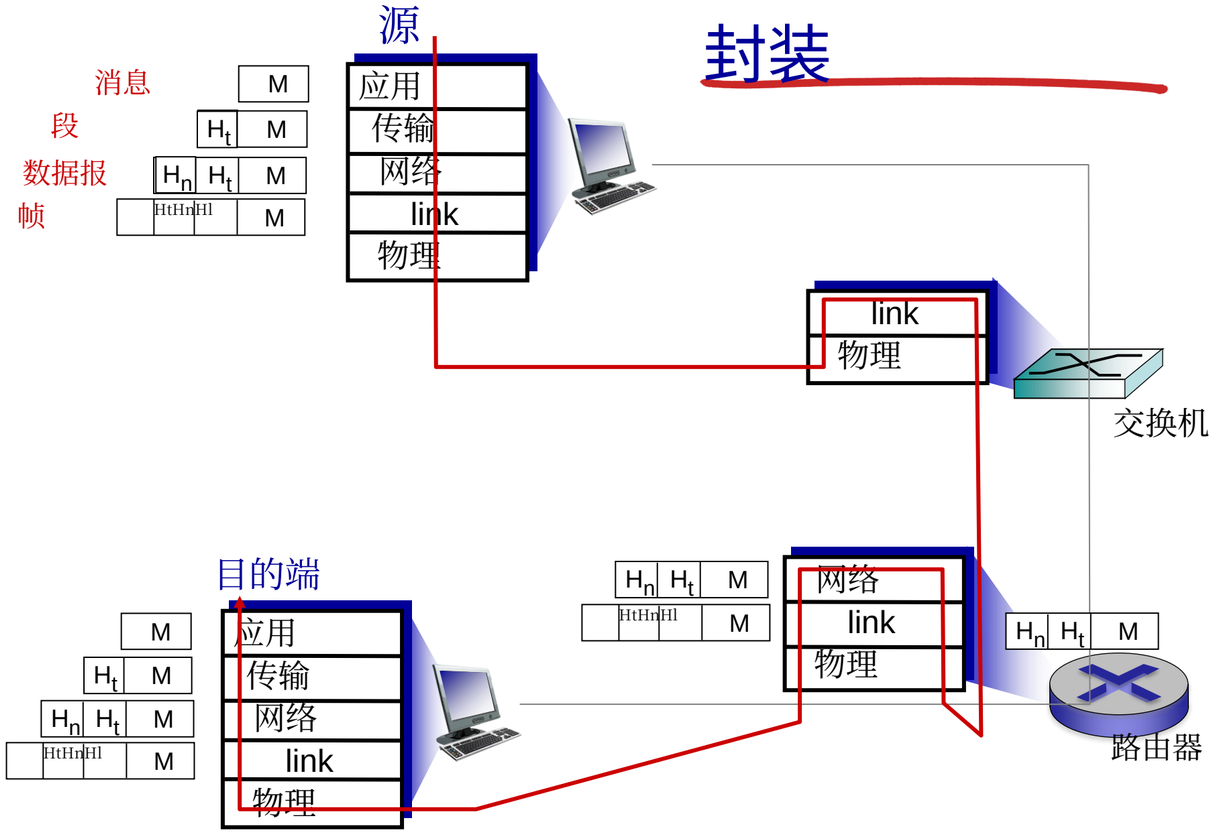
服务原语　　：交互过程中需要交换的命令
在相邻层之间

# Encapsulation

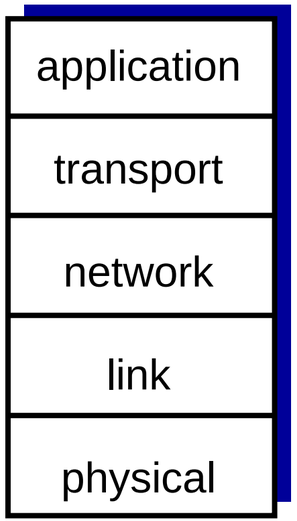


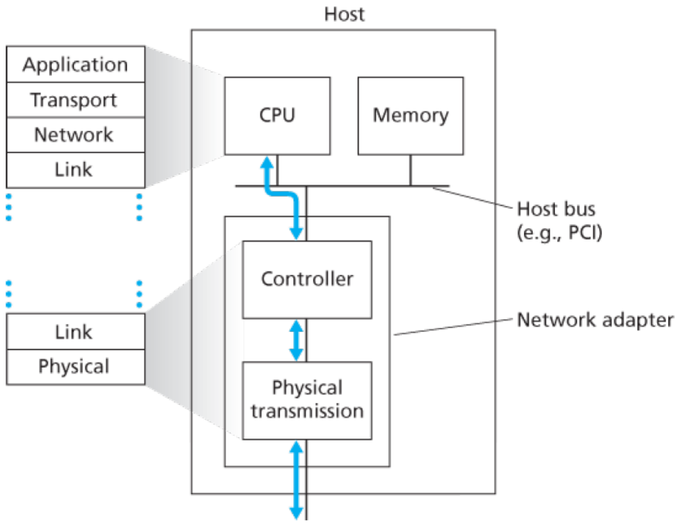



# 封装



# Internet protocol stack

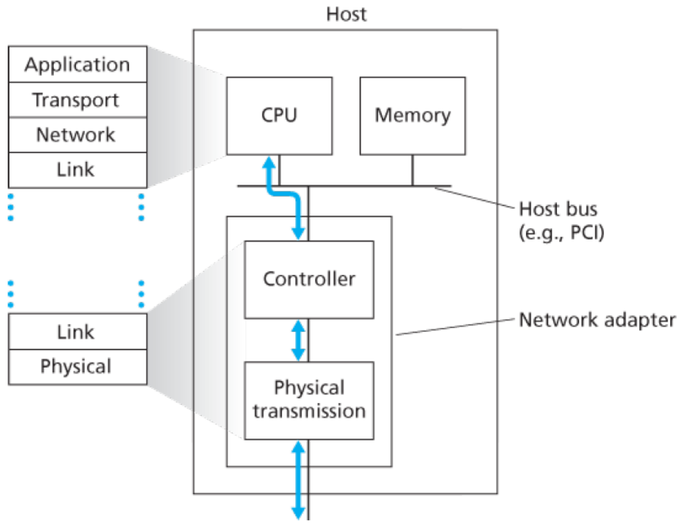





# 互联网协议栈



# Exercise-4

**1. In the OSI model, the first bottom-up level to provide end-to-end services is ( )**

A. data link layer B. transport layer C. session layer D. application layer

**2. Among the following options, which is not described in the network architecture is ( )**

A. layers in the network B. protocols used by each layer
C. internal implementation details of the protocols
D. functions that must be completed by each layer

**3. In the TCP / IP reference model, what directly provides services for the application layer is ( )**

A. presentation layer B. transmission layer C. network layer D. network interface layer

# 练习-4

1. 在OSI模型中，最先从下往上提供端到端服务的是（）

A. 数据链路层 B. 传输层 C. 会话层 D. 应用层

2. 在以下选项中，哪一个没有在网络架构中描述（）

A. 网络中的层 B. 每一层使用的协议 C. 协议的内部实现细节 D. 每一层必须完成的函数

3. 在TCP/IP参考模型中，直接为应用层提供服务的层是（）

A. 表示层 B. 传输层 C. 网络层 D. 网络接口层

# Exercise-4

**4. Assuming that the application layer in the ISO/OSI reference model wants to send 400B data (without splitting), except the physical layer and application layer, all other layers introduce 20B additional overhead when packaging PDU, the transmission efficiency of the application layer is about ( )**

A.80%　　B.83%　　C.87%　　D.91%

# 练习-4

4. 假设 ISO/OSI 参考模型中的应用层要发送 400B 数据（不拆分），除物理层和应用层外，其他层在打包 PDU 时都会引入 20B 的额外开销，应用层的传输效率约为（）

A.80% B.83% C.87% D.91%

# Chapter 1: roadmap

1.1 What is the Internet?
1.2 Network edge
- hosts, access network, physical media

1.3 Network core
- packet switching, circuit switching, network structure

1.4 Delay & loss in packet-switched networks
1.5 Protocol layers, Service models
1.6* Internet history



# 第一章：路线图

1.1 什么是互联网?
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- 主机、接入网、物理介质

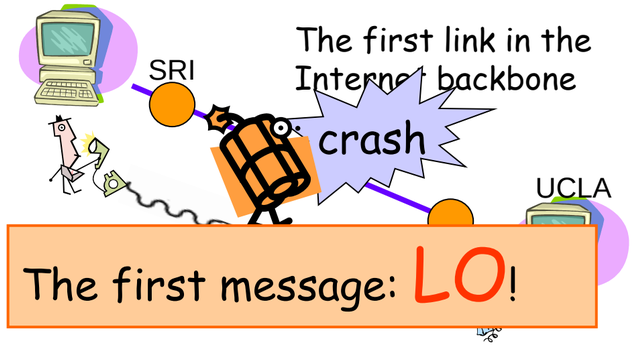
1.3 网络核心
- 分组交换、电路交换、网络结构

1.4 延迟与分组交换网络中的丢失
1.5 协议层与服务模型
1.6* 互联网历史

# Internet History

## *1961-1972: Early packet-switching principles*

- 1961: Kleinrock - queueing theory shows effectiveness of packet-switching
- 1964: Baran - packet-switching in military nets
- 1967: ARPAnet conceived by Advanced Research Projects Agency
- 1969: first ARPAnet node operational




What was the first message ever sent on the Internet? (LOGIN)

- We sent an "L" - did you get the "L"? YEP!
- We sent an "O" - did you get the "O"? YEP!
- We sent a "G" - did you get the "G"?



# 互联网历史

## 1961-1972：早期分组交换原理

- 1961: 克莱因罗克 - 排队论表明分组交换的有效性
- 1964:巴兰 - 军事网络中的分组交换
- 1967: ARPANET 由高级研究计划局构想
- 1969年：第一个ARPANET节点开始运行



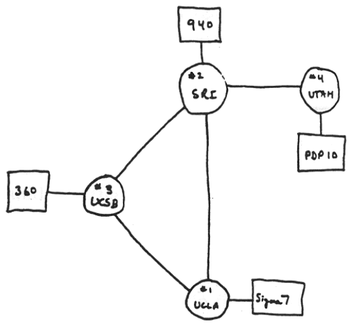

互联网上发送的第一条消息是什么? (LOGIN)

- 我们发送了“L” - 你收到“L”了吗？是的！
- 我们发送了“O” - 你收到“O”了吗？是的！
- 我们发送了“G” - 你收到“G”了吗?

# Internet History

*1961-1972: Early packet-switching principles*

- The Internet is Born! at UCLA on October 29, 1969
- What it looked like at the end of 1969




THE ARPA NETWORK

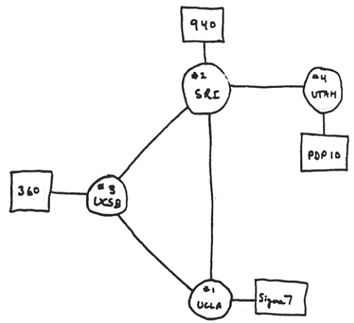
- 1972:
  - ARPAnet public demonstration
  - NCP (Network Control Protocol) first host-host protocol
  - First e-mail program
  - ARPAnet has 15 nodes



# 互联网历史

1961-1972：早期分组交换原理

- 互联网诞生！于1969年10月29日在加州大学洛杉矶分校
- 1969年末时的样子




THE ARPA NETWORK

- 1972:
  - ARPANET 公共演示
  - 网络控制协议（NCP）首个主机-主机协议•首个电子邮件程序• ARPANET有15个节点

# Internet History

## *1972-1980: Internetworking, new and proprietary nets*

- 1970: ALOHAnet satellite network in Hawaii
- 1974: Cerf and Kahn - architecture for interconnecting networks
- 1976: Ethernet at Xerox PARC
- ate70's: proprietary architectures: DECnet, SNA, XNA
- late 70's: switching fixed length packets (ATM precursor)
- 1979: ARPAnet has 200 nodes

Cerf and Kahn's internetworking principles:

- minimalism, autonomy - no internal changes required to interconnect networks
- best effort service model
- stateless routers
- decentralized control

define today's Internet architecture



# 互联网历史

## 1972-1980：互联网互连，新的和专有网络

- 1970：夏威夷的ALOHAnet卫星网络
- 1974:Cerf和Kahn - 互联网互连网络架构
- 1976:以太网在Xerox PARC
- 70年代中期：专有架构：DECnet、SNA、XNA
- 70年代后期：交换固定长度数据包（ATM前身）
- 1979：ARPANET有200个节点

Cerf和Kahn的互联网互连原则：

- 极简主义，自主性 - 无需对内部进行更改即可互连网络
- 尽力而为服务模型
- 无状态路由器
- 分布式控制

定义当今的互联网架构

# Internet History

*1980-1990: new protocols, a proliferation of networks*

- 1983: deployment of TCP/IP
- 1982: SMTP e-mail protocol defined
- 1983: DNS defined for name-to-IP-address translation
- 1985: FTP protocol defined
- 1988: TCP congestion control

- New national networks: Csnet, BITnet, NSFnet, Minitel
- 100,000 hosts connected to confederation of networks



# 互联网历史

1980-1990：新协议，网络激增

- 1983: TCP/IP的部署
- 1982: SMTP电子邮件协议定义
- 1983: DNS定义用于名称到IP地址的转换
- 1985:FTP协议定义
- 1988:TCP拥塞控制

- 新的国家网络：Csnet、BITnet、NSFnet、Minitel
- 10万台主机连接到网络联盟

# Internet History

*1990, 2000's: commercialization, the Web, new apps*

- Early 1990's: ARPAnet decommissioned
- 1991: NSF lifts restrictions on commercial use of NSFnet (decommissioned, 1995)
- Early 1990s: Web
  - Hypertext [Bush 1945, Nelson 1960's]
  - HTML, HTTP: Berners-Lee
  - 1994: Mosaic, later Netscape
  - Late 1990's: commercialization of the Web

Late 1990's – 2000's:

- More killer apps: instant messaging, P2P file sharing
- Network security to forefront
- Est. 50 million host, 100 million+ users
- Backbone links running at Gbps



# 互联网历史

1990，2000 年代：商业化，万维网，新应用

- 20世纪90年代初：ARPANET停止运行
- 1991年： NSF解除对NSFnet商业使用的限制（1995年退役）
- 20世纪90年代初： 万维网
  - 超文本 [布什1945年，尼尔森1960年代]
  - HTML，HTTP：伯纳斯-李
  - 1994: Mosaic，后来成为网景浏览器
  - 1990年代后期：Web的商业化

20世纪90年代末 – 2000年代：

- 更多杀手级应用：即时消息、P2P文件共享
- 网络安全成为焦点
- 估计 5000 万主机，1 亿+ 用户
- 骨干链路运行在 Gbps

# Internet History

## *Recent Developments*





- High-speed access networking
  - 10Gbits Ethernet
  - Wireless access (UWB, WiMAX, 4G, etc.)



- Security
  - Internet
  - Wireless(5G)





- Other networking
  - Industrial internet
  - Internet of things
  - ……



# 互联网历史

## 近期发展





- 高速接入网络
  - 10G以太网 无线接入（UWB、WiMAX、4G等）



- 安全
  - 互联网 无线(5G)





- 其他网络
  - 工业互联网 物联网 ……

# Introduction: Summary

Covered a "ton" of material!

- Internet overview
- What's a protocol?
- Network edge, core, access network
  - Packet-switching versus circuit-switching
- Performance: loss, delay
- Layering and service models
- History

You now have:

- Context, overview, "feel" of networking
- More depth, detail *to follow!*





# 引言：摘要

涵盖了大量的内容！

- 互联网概述
- 什么是协议?
- 网络边缘、核心、接入网
  - 分组交换与电路交换
- 性能：丢失、延迟
- 分层与服务模型
- 历史

您现在拥有：

- 上下文，概述，“网络的感觉”
- 更多深度，细节将随后提供！

